Anurag ENGINEERING COLLEGE
(An Autonomous Institution & ISO 9001:2015 Certified Institute)

Accredited By
A+ Grade

I & II YEAR COURSE STRUCTURE AND DETAILED SYLLABUS

B.Tech

ELECTRICAL AND ELECTRONICS ENGINEERING

R25
Regulation
Under
CBCS



A Four-Year Regular Degree Program
(Applicable for the batches admitted from the Academic Year 2025 – 26)
&
A Three-Year Regular Degree Program under Lateral Entry Scheme (LES)
(Applicable for the batches admitted from the Academic Year 2026 – 27)

ENGINEERING ENGINEERS

ANURAG ENGINEERING COLLEGE

(An Autonomous Institution)

Ananthagiri (V&M), Kodad, Suryapet(Dt), Telangana (India)

ACADEMIC REGULATIONS FOR B. TECH. REGULAR STUDENTS

WITH EFFECT FROM THE ACADEMIC YEAR 2025-26 (R-25)

1.0 Under-Graduate Degree Programme in Engineering & Technology (UGP in E&T)

Anurag Engineering College offers new regulations termed as R-25 regulations for four-year (eight semesters) **Bachelor of Technology** (B.Tech.) degree programme, under Choice Based Credit System (CBCS), with effect from the academic year **2025-26.**

2.0 Eligibility for Admission

2.1 Admissions to the undergraduate (UG) programme shall be made either on the basis of the merit rank obtained by the qualified students at the entrance test conducted by Telangana Government (EAPCET) or the University or on the basis of any other order of merit approved by the University, subject to reservations as prescribed by the government from time to time.

2.2 The medium of instruction for the entire undergraduate programme in Engineering & Technology will be **English** only.

3.0 B.Tech. Programme Structure

3.1 A student after securing admission shall complete the B.Tech. programme in a minimum period of **four** academic years and a maximum period of **eight** academic years starting from the date of commencement of first year first semester, failing which student shall forfeit seat in B.Tech. course. Each student has to secure a minimum of 160 credits out of 164 credits for successful completion of the undergraduate programme and award of the B.Tech. degree.

3.2 **UGC/ AICTE** specified definitions/ descriptions are adopted appropriately for various terms and abbreviations used in these academic regulations/ norms.

3.2.1 Semester Scheme

The undergraduate programme is of four academic years and there shall be two semesters in each academic year. There shall be a minimum of 15 weeks of instruction, excluding the mid-term and semester-end exams. Around 15 instruction hours, 30 instruction hours and 45 hours of learning need to be followed per one credit of theory course, practical course and project/field-based learning respectively. In each semester, there shall be 'Continuous Internal Evaluation (CIE)' and 'Semester End Examination (SEE)' under Choice Based Credit System (CBCS) and credit-based semester system (CBSS) indicated by UGC. The curriculum/course structure suggested by AICTE are followed.

3.2.2 Credit Courses

All courses offered in each semester are to be registered by the student. Against each course in the course structure, the L: T: P: C (lecture periods: tutorial periods: practical periods: credits) pattern has been defined.

- One credit is allocated for one hour per week in a semester for lecture (L) or Tutorial (T) session.
- One credit is allocated for two hours per week in a semester for Laboratory/ Practical (P) session.
- One credit is allocated for three hours per week in a semester for Project/Mini-Project session.

For example, a theory course with three credit weightage requires three hours of classroom instruction per week, totaling approximately 45 hours of instruction over the entire semester.

3.2.3 Subject Course Classification

All subjects/courses offered for the undergraduate programme in E&T (B.Tech. degree programmes) are broadly classified as follows.

S. No.	Broad Course Classification	Course Group/ Category	Course Description
1	Foundation Courses (FnC)	BS – Basic Sciences	Includes Mathematics, Physics and Chemistry courses
2		ES - Engineering Sciences	Includes Fundamental Engineering Courses
3		HS – Humanities and Social Sciences	Includes courses related to Humanities, Social Sciences and Management
4	Core Courses (CoC)	PC – Professional Core	Includes core courses related to the parent branch of Engineering.
5	Elective Courses (EℓC)	PE – Professional Electives	Includes elective courses related to the parent branch of Engineering.
6		OE – Open Electives	Elective courses which include inter-disciplinary courses or courses in an area outside the parent branch of Engineering.
7	Project Core	Project Work	B.Tech. Project Work
7	Other Core Courses (OCC)	Industry Training/ Internship/ Industry Oriented Mini-project/Skill Development Courses	Industry Training/ Internship/ Industry Oriented Mini-Project/Skill Development Courses
8			
9		Seminar	Seminar based on core contents related to parent branch of Engineering.
10	Skill Development Courses (SDC)	-	Courses designed to help individuals gain, improve, or refine specific skills
11	Value Added Courses (VAC)	-	Courses to build professional values, traditional knowledge and sensitization of societal issues

4.0 Mandatory Induction Programme

An induction program of one week duration for the UG students entering the institution, right at the start shall be implemented. Normal classes commence only after the induction programme is conducted. Following activities could be part of the induction programme: i) Physical Activity, ii) Creative Arts, iii) Imparting Universal Human Values, iv) Literary Activities, v) Lectures by Eminent People, vi) Visits to Local Areas and vii) Familiarization to department as well as entire institute and viii) Making students understand Innovative practices at the college premises etc.

5.0 Course Registration

5.1 A faculty advisor / mentor shall be assigned to a group of around 20 students, who will advise the students about the undergraduate programme, its course structure and curriculum, choices/options of the courses, based on their competence, progress, pre-requisites and interest.

5.2 The academic section of the college invites 'registration forms' from students before the beginning of the semester through 'on-line registration', ensuring 'date and time stamping'. The online registration requests for semester courses shall be completed two weeks before the commencement of SEEs (Semester End Examinations) of the preceding semester.

5.3 A student can apply for **on-line** registration, **only after** obtaining the '**written approval**' from faculty advisor/mentor, which should be submitted to the college academic section through the Head of the Department. A copy of it shall be retained with the Head of the Department, faculty advisor/ mentor and the student.

5.4 A student shall register for all the courses offered in a semester as specified in the course structure.

5.5 Course options exercised through **on-line** registration are final and **cannot** be changed; further, alternative choices also will not be considered. However, if the course that has already been listed for registration by the Head of the Department in a semester could not be offered due to any inevitable or unexpected reasons, then the student shall be allowed to have alternative choice either for a new course (subject to offering of such a course), or for another existing course. Such alternative arrangements will be made by the Head of the Department, with due notification and time-framed schedule, within **a week**, but before the commencement of class-work of the semester.

5.6 The Head of the Department / Course Coordinator should review vacant slots in the timetable of each section once in every week or fortnight. The vacant slots in the time-table may be allocated to the subject teachers who could not take classes in proportion to the number of weeks completed from the commencement of the semester.

5.7 Two faculty members may be allocated for the tutorial session of Mathematics-1 course for better interaction/practice and to minimise the failures in the subject.

5.8 **Professional Electives**: The students have to choose six Professional Electives (PE-I to PE-VI) from the six baskets of professional electives given.

Students have the flexibility to choose from the list of professional electives offered by the Institute or opt to register for the equivalent Massive Open Online Courses (MOOCs) as listed from time to time by the Institution.

5.9 **Open Electives**: Students have to choose three Open Electives (OE-I, II & III) from three baskets of Open Electives given by other than the parent department. However, the student can opt for an Open Elective course offered by his/her parent (Own) department, if the student has not studied that course so far. Similarly, Open Elective courses being studied should not match any courses of the forthcoming semesters.

5.10 Provision for Early Registration of MOOCs:

For a professional elective in a semester, students are allowed to register for an equivalent NPTEL/MOOCs course listed from time to time by the Institution one semester in advance. For example, a Professional Elective of III Year II Sem shall be allowed to register under NPTEL/MOOCs platform in III year I Sem.

The credits earned in one semester in advance can be submitted in the subsequent semester for the assessment. The students who have registered in advance in an equivalent NPTEL/MOOCs course and fail to secure any pass grade in the NPTEL/MOOCs course, can register for the regular course offered in the following semester of their course structure.

5.11 **Conversion of Marks Secured in** NPTEL/MOOCs **into Grades**: Marks secured in the internal and external evaluations of a NPTEL/MOOCs course shall be scaled to 40 and 60 marks respectively. The sum of these two components shall be considered as the total marks out of 100. The corresponding grade shall then be determined as per the marks-to-grades conversion rules specified in Clause 10.3.

5.12 NPTEL/MOOCs are allowed only for professional elective courses and for a few Minors & Honors courses

5.13 **Additional learning resources:**

Students are encouraged to acquire additional course-related knowledge by auditing learning resources from NPTEL/MOOCs platforms for each course offered in their course structure. These additional courses are not meant for earning credits but are intended to enhance knowledge. The Institution shall notify such courses from time to time through their portals for the benefit of students. They are categorized into three types: prerequisite, reinforcement, and aspirational. Prerequisite courses help students gain familiarity and provide sufficient background. Reinforcement courses aim to offer different perspectives on learning, while aspirational courses focus on next-level or advanced learning.

6.0 **Rules to offer Elective courses**

6.1 An elective course may be offered to the students, only if a minimum of 25% of class strength opts for it.

6.2 Same elective course for different sections may be offered by different faculty members. The selection of elective course by students will be based on first come first serve and / or CGPA criterion.

6.3 If the number of students registrations are more than the strength of one section, then it is choice of the concerned Department to offer the same course for more than one section based on the resources available in the department.

7.0 **Attendance requirements:**

7.1 A student shall be eligible to appear for the semester-end examinations, if the student acquires a minimum of 75% of aggregate attendance of all the courses for that semester.

7.2 Shortage of attendance in aggregate upto 10% (securing 65% and above but below 75%) in each semester may be condoned by the college academic committee on genuine and valid grounds, based on the student's representation with supporting evidence.

7.3 A stipulated fee shall be payable for condoning of shortage of attendance as notified in the respective college websites.

7.4 **Two hours** of attendance for each theory course shall be considered, if the student appears for the mid-term examination of that course.

7.5 Shortage of attendance below 65% in aggregate shall in **no** case be condoned.

7.6 Students whose shortage of attendance is not condoned in any semester, are not eligible to take their semester-end examinations of that semester. They get detained and will not be promoted to the next semester. Their registration for that semester shall stand cancelled, including internal marks. They may seek re-registration(re-admission) for that semester in the next academic year.

7.7 A student fulfilling the attendance requirement in the present semester shall not be eligible for readmission into the same semester

8.0 Criteria for Earning of Credits in a Course

8.1 A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to each course, if the student secures not less than 35% (21 marks out of 60 marks) in the semester end examinations (SEE), and a minimum of 40% (40 marks out of 100 marks) in the sum total of the CIE (Continuous Internal Evaluation) and SEE (Semester End Examination) taken together; in terms of letter grades, this implies securing **'C'** grade or above in that course.

8.2 A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to Field Based Research Project / Industry Oriented Mini Project / Internship, if the student secures not less than 40% marks (i.e. 40 out of 100 allotted marks) in each of them. The student is deemed to have failed, if he/she (i) does not submit a report on Field-Based Research Project/Industry Oriented Mini Project/ Internship, or (ii) not make a presentation of the same before the evaluation committee as per schedule, or (iii) secures less than 40% marks in Field-Based Research Project / Industry Oriented Mini Project / Internship evaluations.

8.3 A student eligible to appear in the semester-end examination for any course, is absent from it or failed (thereby failing to secure **'C'** grade or above) may re-appear for that course in the supplementary examination as and when it is conducted. In such cases, internal marks assessed in continuous internal evaluation (CIE) earlier for that course will be carried over, and added to the marks obtained in the SEE supplementary/make-up examination. If the student secures sufficient marks for passing, **'C'** grade or above shall be awarded as specified in clause 10.3.

9.0 Distribution of Marks and Evaluation

9.1 The performance of a student in every course (including Value Added Courses and Skill Development Courses, Laboratory/Practical and Project Work) will be evaluated for 100 marks each, with 40 marks allotted for CIE (Continuous Internal Evaluation) and 60 marks for SEE (Semester End-Examination), irrespective of the credits allocated.

9.2 Continuous Internal Evaluation (CIE)

9.2.1 Theory Courses:

For theory courses, during a semester, there shall be two mid-term examinations. Each Mid-Term examination consists of two parts i) **Part – A** for 10 marks, ii) **Part – B** for 20 marks, totaling to 30 marks. 30 marks. Total duration of mid-term examination is two hours.

1. Mid Term Examination for 30 marks:
   a. Part - A : Objective/quiz paper for 10 marks.
   b. Part - B : Descriptive paper for 20 marks.

The objective/quiz paper is set with multiple choice, fill-in the blanks and match the following type of questions for a total of 10 marks.

The descriptive paper shall contain 6 questions out of which, the student has to answer 4 questions, each carrying 5 marks. The **average of the two Mid Term Examinations** shall be taken as the final marks for Mid Term Examination (for 30 marks).

While the first mid-term examination shall be conducted on 50% of the syllabus, the second mid-term examination shall be conducted on the remaining 50% of the syllabus. Questions will be drawn from the mid-term exam syllabus, ensuring uniform coverage of all topics.

The remaining 10 marks of Continuous Internal Evaluation are distributed as follows:

2. Five marks for the assignment for 5 marks. Student shall submit two assignments and the **average of 2 Assignments** each for 5 marks shall be taken. The first assignment should be submitted before the conduct of the first mid-term examination, and the second assignment should be submitted before the conduct of the second mid-term examination.

3. Five marks for the Viva-Voce/PPT/Poster Presentation/ Case Study on a topic in the concerned subject. This assessment shall be completed before II Mid-Term Examination. The Principals shall schedule these sessions in their semester plan.

9.2.2 Engineering Drawing and Computer Aided Drafting Course:

For this course, 20 marks will be allocated for day-to-day assessments conducted during drawing practice sessions, and another 20 marks will be allocated for the mid-term examination. In the mid-term examination, students shall attempt any four out of six given questions. The first mid-term exam will be conducted in the conventional mode using a drawing board, while the second mid-term exam will be conducted using a CAD package.

9.3 A **Computer-Based Test (CBT**) in each course is available for students who either:

1. missed one of the two mid-term examinations due to unavoidable circumstances, or
2. attended both mid-term examinations but wish to improve their internal marks

The CBT will be conducted at the end of the semester and will carry a total of 30 marks. The marks obtained in the CBT will be considered equivalent to those obtained in one mid-term examination. Zero marks will be awarded to students who are absent from the mid-term examination. The average of the best two scores from the three exams (the two mid-term exams and the CBT), combined with other internal assessment components, will constitute the Continuous Internal Improvement (CII) marks for that specific course. CBT exams shall be conducted by the University.

9.4 Semester End Examination for theory courses

9.4.1 Theory Courses:

The semester end examinations (SEE), for theory courses, will be conducted for 60 marks consisting of two parts viz. i) **Part- A** for 10 marks and ii) **Part - B** for 50 marks.

- **Part-A** is compulsory, consists of five short answer questions covering all units of syllabus; each question carries two marks.
- **Part-B** consists of five questions carrying 10 marks each. There shall be two questions asked in the question paper from each unit with either-or choice and the student should answer either of the two questions. The student shall answer one question from each of five units.

9.4.2 Engineering Drawing and Computer Aided Drafting Course:

Question paper consists of five questions carrying 12 marks each. There shall be two questions asked in the question paper from each unit with either-or choice and the student should answer either of the two questions. The student shall answer one question from each of five units.
There shall be no section with short answer questions.

9.4.3 Duration of SEE:

The duration of Semester End Examination of theory and drawing courses is 3 hours.

9.5 Semester End Examination for Practical Courses

For practical courses there shall be a **Continuous Internal Evaluation (CIE)** during the semester for 40 marks and semester-end examination for 60 marks.

The breakup of the continuous internal evaluation for 40 marks is as follows:

1. 10 marks for a write-up on day-to-day experiments in the laboratory (in terms of aim, components/procedure, expected outcome).
2. 10 marks for viva-voce (or) tutorial (or) case study (or) application (or) poster presentation of the course concerned.

3. 10 marks for the internal practical examination conducted by the laboratory teacher concerned.
4. The remaining 10 marks are for Laboratory Record Report/Project and Presentation, which consists of the Design (or) Software / Hardware Model Presentation (or) App Development (or) Prototype submission which shall be evaluated after completion of laboratory course and before semester end practical examination.

The **Semester End Examination** for practical courses shall be conducted with an external examiner and the laboratory course teacher. The external examiner shall be appointed from the inside or outside college and not from a group colleges.

In the Semester End Examination for practical courses held for 3 hours, rubrics of evaluation for 60 marks is as given below:

1. 10 marks for write-up/ programe
2. 15 for experiment/program
3. 15 for evaluation of results or excution
4. 10 marks for presentation on another experiment/program in the same laboratory course and
5. 10 marks for viva-voce on concerned laboratory course.

For any change of experiment, 5 marks will be deducted from the total of 60 marks. If second time change of experiment is requested, another five marks will be deducted from the 60 marks. No third change will be permitted.

9.6 Field-based Research Project:

There shall be a Field-based Research Project in the intervening summer between II-II and III-I Semesters. Students will register for this project immediately after II Year II Semester examinations and pursue it during summer vacation. The Field-based Research Project shall be submitted in a report form and presented before the committee in III year I semester. It shall be evaluated for 100 external marks. The evaluation committee shall consist of an External Examiner, Head of the Department, Supervisor of the Project and a Senior Faculty Member of the department. There shall be no internal marks for Field-based Research Project. Student shall have to earn 40% marks, i.e 40 marks out of 100 marks. The student is deemed to have failed, if he (i) does not submit a report on the Project, or (ii) does not make a presentation of the same before the committee as per schedule, or (iii) secures less than 40% marks in this course.

9.7 Internship/Industry Oriented Mini Project:

There shall be an Internship/Industry Oriented Mini Project in collaboration with an industry from their specialization. Students shall register for this project immediately after III Year II Semester Examinations and pursue it during summer vacation. Internship should be carried out at an organization (or) Industry. The Industry Oriented Mini Project shall be submitted in a report form and presented before the committee in IV Year I Semester before the semester end examination. It shall be evaluated for 100 external marks. The committee consists of an External Examiner, Head of the Department, Supervisor of the Industry Oriented Mini Project/Internship, and a Senior Faculty Member of the Department.

9.7.1 For evaluating industry-oriented mini-projects, it is preferable to appoint an external examiner from the industry, ideally from one of the organizations/ industries with which the institute has established / proposing to establish collaborations.

9.8 UG Project Work:

9.8.1 The UG project work shall be initiated at the beginning of the IV Year II Semester and the duration of the project work is one semester. The student must present in consultation with his/her supervisor, the title, objective and plan of action of his/her Project work to the departmental committee for approval within two weeks from the commencement of IV Year II Semester. Only after obtaining the approval of the departmental committee, the student can

start his/her project work.

9.8.2 Student has to submit project work report at the end of IV Year II Semester. The project work shall be evaluated for 100 marks. Out of which 40 marks and 60 marks are allocated for CIE and External Evaluation respectively.

9.8.3 For internal evaluation, the departmental committee consisting of Head of the Department, Project Supervisor and a Senior Faculty Member shall evaluate the project work for 40 marks. The distribution of marks is as follows:

- Objective(s) of the work done - 05 Marks
- Methodology adopted - 15 Marks
- Results and Discussions - 15 Marks
- Conclusions and Outcomes - 05 Marks

Total - 40 Marks

9.8.4 The External Evaluation shall be conducted by the external examiner for a total of 60 marks.

- the presentation of the work - 20 Marks
- communication skills - 15 Marks
- viva-voce - 25 Marks

Total - 60 Marks

The topics for main Project shall be different from the topic of Industry Oriented Mini Project/ Internship/SDC. The student is deemed to have failed, if he (i) does not submit a report on the Project, or (ii) does not make a presentation of the same before the External Examiner as per schedule, or (iii) secures less than 40% marks in the sum total of the CIE and SEE taken together.

9.8.5 For conducting viva-voce exam of project work, The institute appoints an external examiner. The external examiner may be selected from the list of experts submitted by the HoD.

9.8.6 A student who has failed, may re-appear once for the above evaluation, when it is scheduled again; if student fails in such 'one re-appearance' evaluation also, he/she has to appear for the same in the next subsequent year, as and when it is scheduled.

9.9 Skill Development Courses:

Four Skill Development Courses are included in the Curriculum in II-I, II-II, III-I and III-II semesters. Each Skill Development Course carries one credit. The evaluation pattern will be same as that of a laboratory course including the internal and external assessments.

The objective of Skill Courses is to develop the cognitive skills as well as the psycho-motor skills.

9.10 Value-Added Courses:

The evaluation of Value-Added Courses shall be similar to that of theory courses. However, the scheduling of these mid-term exams and semester-end examinations may not be combined with main-stream examinations. One hour /45 mins proctored mid-term examination shall be conducted in the regular class by the same subject teacher. It should not impact the conduct of other classes on that day.

The scheduling of the semester-end examinations shall also be intimated by the University time to time.

10.0 Grading Procedure

10.1 Absolute grading system is followed for awarding the grades to each course.

10.2 Grades will be awarded to indicate the performance of students in each Theory, Laboratory, Industry-Oriented Mini Project/ Internship/ Skill development course and Project Work. Based on the percentage of marks obtained (Continuous Internal Evaluation plus Semester End

Examination, both taken together) as specified in clause 8 above, a letter grade shall be given as explained in the following clause.

10.3 To measure the performance of a student, a 10-point grading system is followed. The mapping between the percentage of marks secured and the corresponding letter grade is as follows:

Range of % of Marks Secured in a Course	Letter Grade	Grade Points (GP)
Greater than or equal to 90	O (Outstanding)	10
80 and less than 90	A^+ (Excellent)	9
70 and less than 80	A (Very Good)	8
60 and less than 70	B^+ (Good)	7
50 and less than 60	B (Average)	6
40 and less than 50	C (Pass)	5
Below 40	F (FAIL)	0
Absent	Ab	0

10.4 A student shall be declared successful or 'passed' in a semester, if he/she secures **'C'** grade or above in every course (ie GP ≥ 5)

10.5 A student who has obtained an '**F'** grade in any course shall be deemed to have '**failed'** and is required to re-appear for a supplementary exam as and when conducted. In such cases, internal marks in those courses will remain the same as those obtained earlier.

10.6 To a student who has not appeared for an examination in any course, '**Ab'** grade will be allocated in that course, and he/she is deemed to have '**Failed'**. Such student will be required to re-appear for supplementary/make-up exam as and when conducted. The internal marks in those courses will remain the same as those obtained earlier.

10.7 The students earn a Grade Point (G) in each course, on the basis of letter grade secured in that course. Every student who passes a course will receive grade point **GP >= 5** ('C' grade or above).

10.8 The 'Credit Points' (C) are computed by multiplying the grade point with credits for a given course.

Credit Points (C) = Grade Point (G) x Credits

10.9 The Semester Grade Point Average (SGPA) is calculated only when all the courses offered in a semester are cleared by a student. It is calculated by dividing the sum of credit points (ΣCG) secured from all courses registered in a semester, by the total number of credits registered during that semester. SGPA is rounded off to **two** decimal places. SGPA for each semester is thus computed as

$$\textbf{SGPA} = \{ \sum_{i=1}^{N} C_i G_i \} / \{ \sum_{i=1}^{N} C_i \}$$

where 'i' is the course indicator index (considering all courses in a semester), 'N' is the no. of courses registered for the semester (as listed under the course structure of the branch), C_i is the no. of credits allotted to the ith course, and G_i represents the grade points corresponding to the letter grade awarded for that i^{th} course.

10.10 If a student earns more than 160 credits, only the courses corresponding to the best 160 credits shall be considered for the computation of CGPA of B.Tech. degree.

10.11 The Cumulative Grade Point Average (CGPA) is a measure of the overall cumulative performance of a student in all semesters considered for registration. The CGPA is the ratio of the total credit points secured by a student for the courses correspond to best 160 credits out of **all** registered courses in **all** semesters, and the total number of credits correspond to those selected courses. CGPA is rounded off to **two** decimal places. CGPA is thus computed at the end of each semester, from the I year II semester onwards, as per the formula

where '**M'** is the total no. of courses corresponding to the best 160 credits from the courses registered in all eight semesters, 'j' is the course indicator index (takes into account all courses from 1 to 8 semesters), C_j is the no. of credits allotted to the jth course, and G_j represents the grade points (GP) corresponding to the letter grade awarded for that j^{th} course.

Illustration of the Calculation of SGPA:

Course	Credits	Letter Grade	Grade Points	Credit Points
Course 1	4	A	8	4 x 8 = 32
Course 2	3	O	10	3 x 10 = 30
Course 3	3	C	5	3 x 5 = 15
Course 4	3	B	6	3 x 6 = 18
Course 5	3	A	8	3 x 8 = 24
Course 6	2	A+	9	2 x 9 = 18
Course 7	1	C	5	1 x 5 = 5
Course 8	1	O	10	1 x 10 = 10
	20			152

SGPA = 152/20 = 7.6

The CGPA of the entire B.Tech. programme shall be calculated considering the best 160 credits earned by the student.

10.12 For merit ranking or comparison purposes or for any other listing, **only** the '**rounded off'** values of the CGPAs will be used.

10.13 SGPA of a semester will be mentioned in the semester Memorandum of Grades if all courses of that semester are cleared in first attempt. Otherwise, the SGPA shall be mentioned only on the Memorandum of Grades in which sitting he passed his last exam in that semester.

11.0 Declaration of Results and issue of Grade Memo

11.1 While declaring the results, the web-version should display the marks earned by the students with the internal and external marks break-up. However, in the memorandum of grades, the marks need not be shown.

11.2 After the completion of each semester, a certificate of memorandum of grades shall be issued to all the registered students, indicating the letter grades and credits earned. It will show the details of the courses registered (course code, course title, no. of credits), letter grade and credits earned.

12.0 Withholding of Results

12.1 If the student has not paid the fees to the University at any stage, or has dues pending due to any reason whatsoever, or if any case of indiscipline is pending, the result of the student may be withheld, and the student will not be allowed to go into the next higher semester. The award or issue of the degree may also be withheld in such cases.

13.0 Supplementary Examinations:

13.1 At the end of each semester, along with regular semester examinations, supplementary examinations shall be conducted for the students who have back-log subjects.

13.2 Advanced supplementary examinations in IV Year II Semester courses may be conducted for those who failed in any course offered in IV Year II Semester. It may enable the students to receive their B.Tech. provisional certificate at an early date. Advanced supply examinations may be scheduled within one month period after the declaration of the final semester results.

There shall be no supplementary examination in the successive semester. The students who could not secure any pass grade in advance supplementary examinations have to wait for regular series examination of next batch to write their back-log examination.

14.0 Promotion Rules

S.No.	Promotion	Conditions to be Fulfilled
1	First year first semester to first year second semester	Regular course of study of first year first semester and fulfilment of attendance requirement.
2	First year second semester to Second year first semester	(i) Regular course of study of first year second semester and fulfilment of attendance requirement (ii) Must have secured at least 25% of the total credits up to first year second semester from all the relevant regular and supplementary examinations, whether the student takes those examinations or not.
3.	Second year first semester to Second year second semester	Regular course of study of second year first semester and fulfilment of attendance requirement.
4	Second year second semester to Third year first semester	(i) Regular course of study of second year second semester and fulfilment of attendance requirement. (ii) Must have secured at least 25% of the total credits up to second year second semester from all the relevant regular and supplementary examinations, whether the student takes those examinations or not.
5	Third year first semester to Third year second semester	Regular course of study of third year first semester and fulfilment of attendance requirement.
6	Third year second semester to Fourth year first semester	Regular course of study of third year second semester and fulfilment of attendance requirement.
7	Fourth year first semester to Fourth year second semester	Regular course of study of fourth year first semester and fulfilment of attendance requirement.

15.0 Re-admission after Detention

i) A student detained due to lack of credits, shall be promoted to the next academic year only after acquiring the required number of credits.

ii) A student detained due to shortage of attendance shall be admitted in the same semester in the successive academic years.

iii) When a student is readmitted in the following academic years, the academic regulations under which the student seeks re-admission shall only be applicable to this student, not the academic regulations in which he got admitted in his/her first year of study.

16.0 Credit Exemption

A student (i) shall register for all courses covering 164 credits as specified and listed in the course structure and (ii) earn 160 or more credits to successfully complete the undergraduate programme.

- Best 160 credits shall be considered for CGPA computation. The student can avail exemption of courses ***totaling up to 4 credits*** other than Professional core courses, Laboratory Courses, Seminars, Project Work and Field Based Research Project / Industry Oriented Mini Project / Internship, for optional drop out from these 164 credits registered;
- The semester grade point average (SGPA) of each semester shall be mentioned at the bottom of the grade card, when all the subjects in that semester have been passed by the student.
- Credits earned by the student in either a Minor or Honors program cannot be counted towards the required 160 credits for the award of the B.Tech. degree

17.0 Award of Degree

17.1 A student who registers for all the courses specified in the course structure and secures the required number of 160 credits within 8 academic years from the date of commencement of the first academic year, shall be declared to have qualified for the award of B.Tech. degree in the branch of Engineering selected at the time of admission.

17.2 A student who qualifies for award of the degree as listed in item 17.1 shall be placed in the following classes.

17.3 A student with final CGPA (at the end of the undergraduate programme) ≥ 7.5, and fulfilling the following conditions - shall be placed in '**First Class with Distinction'**:

(i) Should have passed all the courses in '**First Appearance'**

(ii) Should not have been detained or prevented from writing the semester end examinations in any semester due to shortage of attendance or any other reason.

A student not fulfilling any of the above conditions with final CGPA ≥ 7.5 shall be

placed in **'First Class'.**

17.4 Students with final CGPA (at the end of the undergraduate programme) >= 6.5 but < 7.5 shall be placed in **'First Class'.**

17.5 Students with final CGPA (at the end of the undergraduate programme) >= 5.5 but < 6.5, shall be placed in '**Second Class'**.

17.6 All other students who qualify for the award of the degree (as per item 17.1), with final CGPA (at the end of the undergraduate programme) >= 5.00 but < 5.5, shall be placed in '**pass class**'.

17.7 Grace Marks

Grace marks shall be given to those students who complete the course work of four year B. Tech. degree, not secured pass grade in not more than three subjects and adding a specified grace marks enables the student to pass the subject(s) as well as gets eligibility to receive the provisional degree certificate.

Grace marks for students admitted under the R-25 Academic Regulations should not exceed **0.15%** of the total maximum marks in all eight semesters (excluding the marks allocated for value added courses and skill development courses).

18.0 Award of Gold Medals

18.1 Students fulfilling the conditions listed under item 17.3 alone will be eligible for award of '**Gold Medal**'.

18.2 If more than one student secures the same highest CGPA, then the following tie resolution criteria, in the same order of preference shall be followed for selecting the Gold Medal winner,

until the tie is resolved: 1) more number of times secured highest SGPAs, ii) more number of

O and A+ grades in that order and iii) highest SGPA in the order of first semester to eight semester.

19.0 Conversion of CGPA into equivalent Percentage of Marks

19.1 The following formula shall be used for the conversion of CGPA into equivalent marks, whenever it is necessary

Percentage (%) of Marks = (Final CGPA – 0.5) x 10

20.0 Transitory Regulations for the students re-admitted in R-25 Regulations:

20.1 Transitory regulations are applicable to the students detained due to shortage of attendance as well as detained due to the shortage of credits and seek permission to re-join the B.Tech. programme, where R-25 regulations are in force.

20.2 A student detained due to shortage of attendance and re-admitted in R-25 regulations: Such students shall be permitted to join the same semester, but in R-25 Regulations

20.3 A student detained due to shortage of credits and re-admitted in R-25 regulations: Such students shall be promoted to the next semester in R-25 regulations, only after acquiring the required number of credits as per the corresponding regulations of his/her previous semester.

20.4 A student who has failed in any course in a specific regulation has to pass those courses in the same regulations.

20.5 If a student is readmitted to R-25 Regulations and has any course with 80% of syllabus common with his/her previous regulations, that particular course in R-25 Regulations will be substituted by an equivalent course of R-22 regulations by the Institute . All these details are summarized in a set of look-up Table; one set for each B. Tech. branch.

20.6 Look Up Table of equivalence courses

20.6.1 A lookup table will be provided for the benefit of students and Principals. This lookup table will include all the courses to be registered by students who have been re-admitted under the R-25 Academic Regulations from the R-22 Academic Regulations. Separate lookup tables will be provided for the following categories of students:

1. Students re-admitted into the I Year II Semester of the R-25 Regulations
2. Students re-admitted into the II Year I Semester of the R-25 Regulations
3. Students re-admitted into the II Year II Semester of the R-25 Regulations
4. Students re-admitted into the III Year I Semester of the R-25 Regulations
5. Students re-admitted into the III Year II Semester of the R-25 Regulations
6. Students re-admitted into the IV Year I Semester of the R-25 Regulations
7. Students re-admitted into the IV Year II Semester of the R-25 Regulations

For every B.Tech. branch there shall be separate set of seven lookup tables.

20.6.2 Applicability of Look-up Table: The above look-up table shall be applicable for i) students who seek re admission from R-22 regulations to R-25 regulation and are going to be re-admitted in the same JNTUH affiliated autonomous college and ii) detained students of one JNTUH affiliated autonomous college who seek admission into another JNTUH affiliated autonomous college.

For the above category of students, the Principals of the affiliated colleges need not consult the University for the equivalence courses. However the Principals need to inform in the specified format, the list of such students and equivalences derived from the transitory regulations.

20.6.3 These look-Up Tables are not applicable for the students who seek transfer from i) other Universities to JNTUH affiliated autonomous colleges, ii) autonomous to non-autonomous colleges, III) one autonomous to another autonomous colleges and iv) non-autonomous to

autonomous colleges under JNTUH. Such students should consult the University regarding equivalent courses, as was in previous practice.

20.7 The R-25 Academic Regulations are applicable to a student from the year of re-admission.

However, the student is required to complete the study of B.Tech. degree within the stipulated period of eight academic years from the year of first admission.

21.0 Student Transfers

21.1 There shall be no branch transfers after the completion of admission process.

21.2 There shall be no transfers from one college to another within the constituent colleges and units of Jawaharlal Nehru Technological University Hyderabad.

21.3 The students seeking transfer to colleges affiliated to JNTUH from various other Universities/institutions is having back-logs at the previous University/institute, have to pass the courses offered at JNTUH which are equivalent to the failed courses at the previous University/institute.

21.4 The transferred students from other Universities/Institutions to JNTUH affiliated colleges, shall be given a chance to write CBTs for getting CIE component in the **equivalent course(s)** as per the clearance letter issued by the University.

22.0 Value Added Courses

22.1 Faculty members who have received a certificate in Innovation and Entrepreneurship / Entrepreneurship from a reputed foundation/organization may be given preference to teach the "Innovation and Entrepreneurship" course. This certificate course should include an assessment. Total training duration (online or physical), excluding assessment, should be at least 30 hours. Faculty members from all disciplines with innovative mindset and aptitude to co-create an entrepreneurial ecosystem are eligible to teach this subject.

22.2 Faculty members who have credited a course on Intellectual Property Rights in their UG or PG programme or credited an equivalent course in MOOCs platform/ reputed foundation/ organization in which assessment is a part, may be given preference to teach the elective course on Intellectual Property Rights.

22.3 To ensure quality delivery and standardization in teaching the **Indian Knowledge System (IKS)** and other value-added courses, the following guidelines must be adhered to: i) faculty members must undergo a Faculty Development Program (FDP) organized by UGC-MMTTC

(Malaviya Mission Teacher Training Centre), **or** Any other recognized and competent institution/organization offering similar certified programs, ii) the total instructional duration of the FDP should be a around 32 hours or more, III) all sessions in the FDP must be conducted by certified and qualified resource persons with recognized expertise in the respective domains, iv) A formal assessment component must be included as part of the FDP.

23.0 Mapping with the Sustainable Development Goals

All the courses specified in the course structure of every programme are mapped with the one or more sustainable development goals.

24.0 Scope

24.1 The academic regulations should be read as a whole, for the purpose of any interpretation.

24.2 In case of any doubt or ambiguity in the interpretation of the above rules, the decision of the Principal is final.

24.3 The institutioin may change or amend the academic regulations, course structure or syllabi at any time, and the changes or amendments made shall be applicable to all students with effect from the dates notified by the principal.

24.4 Where the words “he”, “him”, “his”, occur in the regulations, they include “she”, “her”, “hers”.

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ANURAG ENGINEERING COLLEGE

(An Autonomous Institution)

Ananthagiri (V&M), Kodad, Suryapet(Dt), Telangana (India)

ACADEMIC REGULATIONS FOR B.TECH. (LATERAL ENTRY SCHEME) FROM THE AY 2026-27

Eligibility for the award of B.Tech. Degree (LES)

1. The LES students after securing admission shall pursue a course of study for not less than three academic years and not more than six academic years.
2. The student shall register for 123/124 credits and secure 120 credits with CGPA ≥ 5 from II year to IV-year B.Tech. programme (LES) for the award of B.Tech. degree.
3. The student can avail exemption of courses ***totaling up to 3/4 credits*** other than Professional core courses, Laboratory Courses, Seminars, Project Work and Field Based Research Project / Industry Oriented Mini Project / Internship, for optional drop out.
4. The students, who fail to fulfil the requirement for the award of the degree in six academic years from the year of admission, shall forfeit their seat in B.Tech.
5. The attendance requirements of B.Tech. (Regular) shall be applicable to B.Tech. (LES).
6. **Promotion rule**

S. No	Promotion	Conditions to be fulfilled
1	Second year first semester to Second year second semester	Regular course of study of second year first semester and fulfilment of attendance requirement.
2	Second year second semester to Third year first semester	(i) Regular course of study of second year second semester and fulfilment of attendance requirement. (ii) Must have secured at least 25% of the total credits up to second year second semester from all the relevant regular and supplementary examinations, whether the student takes those examinations or not.
3	Third year first semester to Third year second semester	Regular course of study of third year first semester and fulfilment of attendance requirement.
4	Third year second semester to Fourth year first semester	Regular course of study of third year second semester and fulfilment of attendance requirement.
5	Fourth year first semester to Fourth year second semester	Regular course of study of fourth year first semester and fulfilment of attendance requirement.

7. All the other regulations as applicable to B.Tech. 4-year degree course (Regular) will hold good for B. Tech. (Lateral Entry Scheme).
8. LES students are not permitted to exit the B.Tech. program after completion of second year (B.Tech. II Year II Semester).

List of PEO's, PSO's & PO's

PROGRAM EDUCATIONAL OBJECTIVES (PEO)

PEO 1: To prepare students to excel in technical profession/industry and/or higher education by acquiring knowledge in mathematics, science and engineering principles.

PEO 2: Able to formulate, analyze, design and create novel products and solutions to electrical and electronics engineering problems those are economically feasible and socially acceptable.

PEO 3: Able to adopt multi-disciplinary environments, leadership qualities, effective communication, professional ethics and lifelong learning process.

PROGRAM SPECIFIC OUTCOMES (PSO)

PSO 1: An ability to demonstrate the knowledge, skill to analyze the cause and effects on electrical system, processes and systems.

PSO 2: Apply the appropriate techniques and modern engineering hardware and software tools in electrical engineering to engage in life-long learning and to successfully adapt in multi-disciplinary environments.

PROGRAM OUTCOMES (PO)

PO 1: An ability to apply the knowledge of mathematics, science and engineering fundamentals.

PO 2: An ability to conduct Investigations using design of experiments, analysis and interpretation of data to arrive at valid conclusions.

PO 3: An ability to design Electrical and Electronics Engineering components and processes within economic, environmental, ethical and manufacturability constraints.

PO 4: An ability to function effectively in multidisciplinary teams.

PO 5: An ability to identify, formulate, analyze and solve Electrical and Electronics Engineering problems.

PO 6: An ability to understand professional, ethical and social responsibility.

PO 7: An ability to communicate effectively through written reports or oral presentations.

PO 8: The broad education necessary to understand the impact of engineering solutions in

a global, economic, environmental, and societal context.

PO 9: An ability to recognize the need and to engage in independent and life-long learning.

PO 10: Knowledge on contemporary issues.

PO 11: An ability to use the appropriate techniques and modern engineering tools necessary for engineering practice.

PO 12: An ability to demonstrate knowledge and understanding of engineering and management principles and apply these to manage projects.

I Year I Semester **Course Structure**

S.No	Course Code	Course Title	L	T	P	Credits
1	2599101	Matrices and Calculus	3	1	0	4
2	2599106	Engineering Physics	3	0	0	3
3	2505103	C Programming & Data Structures	3	0	0	3
4	2503101	Computer Aided Engineering Graphics	2	0	2	3
5	2502101	Electrical Circuits - I	3	0	0	3
6	2599107	Engineering Physics Laboratory	0	0	2	1
7	2505104	C Programming & Data Structures Laboratory	0	0	2	1
8	2502102	Electrical Circuits - I Laboratory	0	0	2	1
9	2505105	IT Workshop	0	0	2	1
10		Induction Program				
		Total Credits	14	1	10	20

I Year II Semester **Course Structure**

S.No	Course Code	Course Title	L	T	P	Credits
1	2599201	Ordinary Differential Equations and Vector Calculus	3	0	0	3
2	2599204	Engineering Chemistry	3	0	0	3
3	2599205	English for Skill Enhancement	3	0	0	3
4	2505201	Python Programming	3	0	0	3
5	2502203	Electrical Circuits - II	3	0	0	3
6	2599206	Engineering Chemistry Laboratory	0	0	2	1
7	2505202	Python Programming Laboratory	0	0	2	1
8	2502204	Electrical Circuits - II Laboratory	0	0	2	1
9	2599207	English Language and Communication Skills Laboratory	0	0	2	1
10	2503202	Engineering Workshop	0	0	2	1
		Total Credits	15	0	10	20

L- Lectures **T - Tutorial** **P – Practical**

II Year I Semester **Course Structure**

S.No	Course Code	Course Title	L	T	P	Credits
1	2502301	Electromagnetic Fields	3	0	0	3
2	2502302	Electrical Machines - I	3	0	0	3
3	2504301	Electronic Devices and Circuits	3	0	0	3
4	2502303	Power Systems - I	3	0	0	3
5	2502304	Electrical Measurements and Sensors	2	0	0	2
6	2588301	Innovation and Entrepreneurship	2	0	0	2
7	2502305	Electrical Machines - I Laboratory	0	0	2	1
8	2502306	Electrical Measurements and Sensors Laboratory	0	0	2	1
9	2504302	Electronic Devices and Circuits Laboratory	0	0	2	1
10	2502307	Design of Electrical Systems using AutoCAD	0	0	2	1
11	2599304	Environmental Science /IKS	1	0	0	1
		Total Credits	18	0	8	21

II Year II Semester **Course Structure**

S.No	Course Code	Course Title	L	T	P	Credits
1	2599402	Numerical Methods and Complex Variables	3	0	0	3
2	2502401	Electrical Machines - II	3	0	0	3
3	2502402	Power Systems - II	3	0	0	3
4	2504401	Digital Electronics	3	0	0	3
5	2502403	Control Systems	3	0	0	3
6	2599403	Computational Mathematics Laboratory (Using Python/MATLAB Software)	0	0	2	1
7	2502404	Electrical Machines - II Laboratory	0	0	2	1
8	2502405	Control Systems Laboratory	0	0	2	1
9	2504402	Digital Electronics Laboratory	0	0	2	1
10	2502406	PCB Design	0	0	2	1
11	2599404	Gender Sensitization Lab/ Human Values and Professional Ethics	1	0	0	1
		Total Credits	16	0	10	21

L- Lectures **T - Tutorial** **P – Practical**

I Year I Semester

I B.Tech. I Semester

Course Name	:	**MATRICES AND CALCULUS**				
Course Code	:	**2599101**	L	T	P	C
Course Category	:	**Theory**	3	1	0	4

CIE: 40 SEE: 60 Total: 100

Course Objectives: To learn

- Applying basic operations on matrices and their properties. Concept of a rank of the matrix and applying this concept to know solving the system of linear equations.
- Concept of eigen values and eigen vectors and to reduce the quadratic form to canonical form
- Concept of the mean value theorems and their application to the mathematical problems
- Finding maxima and minima of functions of three variables.
- Evaluation of multiple integrals and their applications.

UNIT-I: Matrices **8 Hrs**

Rank of a matrix by Echelon form and Normal form – Inverse of Non-singular matrices by Gauss-Jordan method.

System of linear equations: Solving system of Non-Homogeneous equations by Gauss elimination method and Gauss Seidel Iteration Method.

UNIT-II: Eigen values and Eigen vectors **10 Hrs**

Linear Transformation and Orthogonal Transformation: Cayley-Hamilton Theorem (without proof), Eigen values – Eigen vectors and their properties – Diagonalization of a matrix.

Quadratic forms and Nature of the Quadratic Forms – Reduction of Quadratic form to canonical form by Orthogonal Transformation.

UNIT-III: Single Variable Calculus **10 Hrs**

Limit and Continuous of functions and its properties. Mean value theorems: Rolle's theorem – Lagrange's Mean value theorem with applications – Cauchy's Mean value Theorem-Taylor's Series – Maclaurin's Series (All the theorems without proof).

Curve Tracing: Curve tracing in cartesian coordinates.

UNIT-IV: Multivariable Calculus (Partial Differentiation and applications) **10 Hrs**

Definitions of Limit and continuity – Partial Differentiation: Total derivative – Jacobian – Functional dependence & independence.

Applications: Maxima and minima of functions of three variables using method of Lagrange multipliers.

UNIT-V: Multivariable Calculus (Integration) **10 Hrs**

Evaluation of Double Integrals (Cartesian and polar coordinates) – change of order of integration (only Cartesian form) – Change of variables for double integrals (Cartesian to polar). Evaluation of Triple Integrals

Applications: Areas by double integrals and volumes by triple integrals.

TEXT BOOKS:

1. B.S. Grewal, Higher Engineering Mathematics, Khanna Publishers, 36th Edition, 2010.
2. R.K. Jain and S.R.K. Iyengar, Advanced Engineering Mathematics, Narosa Publications, 5th Editon, 2016.

REFERENCES:

1. Erwin Kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006.
2. G.B. Thomas and R.L. Finney, Calculus and Analytic geometry, 9thEdition, Pearson, Reprint, 2002.
3. N.P. Bali and Manish Goyal, A text book of Engineering Mathematics, Laxmi Publications, Reprint, 2008.
4. H. K. Dass and Er. Rajnish Verma, Higher Engineering Mathematics, S Chand and Company Limited, New Delhi.

Course Outcomes: After learning the contents of this paper, the student must be able to

CO-1. Write the matrix representation of a set of linear equations and to analyze the solution of the system of equations

CO-2. Find the Eigen values and Eigen vectors. Reduce the quadratic form to canonical form using orthogonal transformations.

CO-3. Solve the applications of the mean value theorems.

CO-4. Find the extreme values of functions of Three variables with/ without constraints.

CO-5. Evaluate the multiple integrals and apply the concept to find areas, volumes.

Course Articulation Matrix:

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**1**	**2**
CO-1	3	2	3	2	3	2	2	2	-	-	2	2	2	2
CO-2	3	2	3	2	3	2	2	2	-	-	2	2	2	3
CO-3	3	2	3	2	3	2	2	2	-	-	2	2	2	3
CO-4	3	2	3	2	3	2	2	2	-	-	2	2	3	3
CO-5	3	2	3	2	3	2	2	2	-	-	2	2	3	3

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

I B.Tech. I Semester

Course Name	:	**ENGINEERING PHYSICS**				
Course Code	:	**2599106**	L	T	P	C
Course Category	:	**Theory**	3	0	0	3

CIE: 40 SEE: 60 Total: 100

Course Objectives:

- To understand fundamental concepts of quantum mechanics and their applications in solids and nanomaterials.
- To give students a clear understanding of how semiconductors and nanomaterials work, how they are made, and how they are used in modern technology
- To introduce quantum computing principles, quantum gates, and basic quantum algorithms.
- To learn the properties and applications of magnetic and dielectric materials.
- To explore the working and applications of lasers and fibre optics in modern technology.

UNIT-I: 12 Hrs

Quantum Mechanics: Introduction, de-Broglie hypothesis, Davisson-Germer Experiment Heisenberg uncertainty principle, physical significance of wave function, postulates of quantum mechanics (Elementary treatment): operators in quantum mechanics, eigen values and eigen functions, expectation value; Schrödinger's time independent wave equation, particle in a 1D box.

Bloch's theorem (Elementary treatment), Kronig-Penney model (Elementary treatment): E-K diagram, effective mass of electron (Elementary treatment), formation of energy bands, classification of solids.

UNIT-II: 12 Hrs

Semiconductor Physics: Intrinsic and extrinsic semiconductors, variation of Fermi energy level with temperature and doping concentration (Elementary treatment), direct and indirect band gap semiconductors, Hall Effect, construction, principle, V-I characteristics and applications of P-N Junction diode, LED and solar cell.

Nanotechnology: Introduction to nanoscience and nanotechnology, Classification of nanomaterials: 0D,1D,2D and 3D, surface to volume ratio, quantum confinement, synthesis methods: Ball milling, sol-gel method, Characterization Techniques: X-ray diffraction (XRD) and Scanning Electron Microscopy (SEM), applications of nanomaterials.

UNIT-III: 12Hrs

Quantum Computing: Introduction, linear algebra for quantum computation(Elementary treatment), Dirac's Bra and Ket notation and their properties, superposition, entanglement, Hilbert space, Bloch's sphere, concept of quantum computer: classical bits, Qubits, multiple Qubit system, quantum computing system for information processing (Elementary treatment), evolution of quantum systems, quantum measurements, quantum gates (Elementary treatment): Pauli, Hadamard, and CNOT gates, challenges and advantages of quantum computing over classical computation.

UNIT-IV: 12 Hrs

Dielectric materials: Introduction to dielectric materials, types of polarization: electronic, ionic (Quantitative) & orientation (Elementary treatment); ferroelectric, piezoelectric, pyroelectric materials and their properties and applications

Magnetic materials: Introduction to magnetic materials, origin of magnetic moment- classification of magnetic materials: dia, para, ferro, anti-ferro and ferri magnetic materials. Weiss domain theory of ferromagnetism, Hysteresis and B-H curve, soft and hard magnetic materials, applications.

UNIT-V: **12 Hrs**

Laser: Introduction to laser, characteristics of laser, Einstein coefficients and their relations, metastable state, population inversion, pumping, lasing action, He-Ne laser, Nd-YAG Laser, semiconductor diode laser, applications of Lasers.

Fibre Optics: Introduction to fibre optics, total internal reflection, Physical components of optical fibre, acceptance angle, numerical aperture, classification of optical fibres (based on refractive index profile), losses in optical fibre, applications: optical fibre for communication system, sensor for structural health monitoring.

Textbooks:

1. M.N Avadhanulu, P.G Kshirsagar, T VS Arun Murthy, A Textbook of Engineering Physics, S Chand,2019.
2. Shatendra Sharma and Jyotsna Sharma, Engineering Physics, Pearson India Education Services Pvt. Ltd,2018.
3. Michael A. Nielsen & Isaac L. Chuang, *Quantum Computation and Quantum Information*, Cambridge University Press.

References:

1. Charles Kittel, *Introduction to Solid State Physics*, John Wiley & Sons, Inc.
2. Md. N. Khan, S. Panigrahi, Principles of Engineering Physics 1 & 2, Cambridge University Press, 2016
3. Prof. A. K. Katiyar, Dr. C. K. Pandey, Engineering Physics: Theory and Practical, Wiley India Pvt. Ltd., India,2017.
4. Halliday, Resnick and Walker, Fundamentals of Physics, John Wiley &Sons,11th Edition, 2018.

Course Outcomes:

CO1: Apply quantum mechanical principles to explain particle behaviour and energy band formation in solids

CO2: Able to explain the principles, types, fabrication methods, and applications of semiconductors and nanomaterials in simple terms.

CO3: Understand quantum computing concepts, use quantum gates, and explain basic quantum algorithms.

CO4: Classify magnetic and dielectric materials and explain their properties, synthesis, and applications.

CO5: Explain the principles of lasers and fibre optics and their applications in communication and sensing.

Course Articulation Matrix:

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO-1	2	3	2	1	-	-	-	-	-	-	-	2	2	2
CO-2	2	2	-	1	-	-	-	-	-	-	-	1	2	1
CO-3	2	2	-	2	-	-	-	-	-	-	-	1	2	1
CO-4	2	2	1	1	-	-	-	-	-	-	-	-	2	2
CO-5	2	3	-	2	-	-	-	-	-	-	-	1	1	2

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

I B.Tech. I Semester

Course Name	:	**C PROGRAMMING AND DATA STRUCTURES**				
Course Code	:	**2505103**	L	T	P	C
Course Category	:	**Theory**	3	0	0	3

CIE: 40 SEE: 60 Total: 100

Course Objectives:

- To introduce computer systems, environments, programming basics, and the fundamentals of C language.
- To understand structured programming concepts using statements, functions, and recursion.
- To apply arrays, pointers, and searching/sorting techniques for computational problem solving.
- To develop applications using strings, structures, and derived data types.
- To implement fundamental data structures (linked lists, stacks, and queues) for problem-solving.

UNIT-I: **12 Hrs**

Introduction to Computers – Computer Systems, Computing Environments, Computer Languages, Creating and running programs

Introduction to C Language – Background, Simple C programs, Identifiers, Basic data types, Variables, Constants, Input / Output, Structure of a C Program – Operators, Bit-wise operators, Expressions Expression Evaluation, Type conversions.

UNIT-II: **12 Hrs**

Statements – if and switch statements, Repetition statements – while, for, do-while statements, other statements related to looping – break, continue, go to, Designing Structured Programs- Functions, basics, user defined functions, inter function communication, standard functions, Recursion.

UNIT-III: **12Hrs**

Arrays – Concepts, using arrays in C, two –dimensional arrays, multi-dimensional arrays. Searching and sorting

Pointers – Introduction, Pointers for inter function communication, pointers to pointers, compatibility, Pointer Applications – Passing an array to a function.

UNIT-IV: **12 Hrs**

Strings – Concepts, C Strings, String Input / Output functions, arrays of strings, string manipulation functions, string / data conversion.

Derived types – Structures – Declaration, definition and initialization of structures, accessing structures, operations on structures, nested structures, The Type def, enumerated types

UNIT-V: **12 Hrs**

Data Structures – Introduction to Data Structures, abstract data types, Linear list – singly linked list implementation, insertion, deletion and searching operations on linear list, Stacks-Operations, array and linked representations of stacks, stack applications, Queues-operations, array and linked representations.

Textbooks:

1. C Programming & Data Structures, B.A.Forouzan and R.F. Gilberg, Third Edition, Cengage Learning.
2. Problem Solving and Program Design in C, J.R. Hanly and E.B. Koffman, Fifth Edition, Pearson Education.
3. The C Programming Language, B.W. Kernighan and Dennis M.Ritchie, PHI/Pearson Education

References:

After completion of the Course, the student should be able to

1. C & Data structures – P. Padmanabham, 3rd Edition, B.S. Publications.
2. C Programming with problem solving, J.A. Jones & K. Harrow, Dreamtech Press
3. Programming in C – Stephen G. Kochan, III Edition, Pearson Education.
4. C for Engineers and Scientists, H. Cheng, McGraw-Hill International Edition
5. Data Structures using C – A. M. Tanenbaum, Y. Langsam, and M.J. Augenstein, Pearson Education / PHI
6. C Programming & Data Structures, E. Balagurusamy, TMH.
7. C Programming & Data Structures, P. Dey, M Ghosh R Thereja, Oxford University Press
8. C & Data structures – E V Prasad and N B Venkateswarlu, S. Chand & Co.

Course Outcomes:
After completion of the Course, the student should be able to

CO-1. Understand computer systems, environments, C syntax, operators, and simple programs.
CO-2. Apply structured programming constructs, functions, and recursion to solve problems.
CO-3. Use arrays, pointers, searching and sorting techniques for solving computational tasks.
CO-4. Develop programs using strings, structures, and derived data types.
CO-5. Implement linear data structures (linked lists, stacks, queues) to solve engineering problems.

Course Articulation Matrix:

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**1**	**2**
CO-1	3	2	1	-	1	-	-	-	-	1	-	-	3	1
CO-2	3	3	2	-	2	-	-	-	-	1	-	-	3	2
CO-3	2	3	3	-	3	1	-	-	-	2	1	-	3	2
CO-4	2	2	2	-	2	1	-	-	-	2	2	-	2	2
CO-5	2	3	3	-	3	2	-	1	-	2	2	-	3	3

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

I B.Tech. I Semester

Course Name	:	**COMPUTER AIDED ENGINEERING GRAPHICS**				
Course Code	:	**2503101**	L	T	P	C
Course Category	:	**Theory**	2	0	2	3

CIE: 40 SEE: 60 Total: 100

Course Objectives:

- To introduce the fundamentals of engineering drawing and projection systems.
- To develop skills in constructing orthographic, isometric, and sectional views.
- To train students in interpreting and creating technical drawings using CAD tools.
- To familiarize students with dimensioning standards and drafting conventions.
- To bridge manual drafting techniques with computer-aided drafting practices.

UNIT-I: **13Hrs**

Introduction to Engineering Graphics (Conventional):

Principles of Engineering Graphics and their Significance, Geometrical Constructions, Scales, Plain and Diagonal, Conic Sections including the Rectangular Hyperbola, General method only. Cycloid, Epicycloids and Hypocycloid.

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UNIT-II: **12Hrs**

Orthographic Projections (Conventional and Computer Aided)

Principles of Orthographic Projections, Conventions, Projections of Points and Lines, Projections of Plane regular geometric figures. Computer aided orthographic projections, points, lines and planes. Introduction to Computer aided drafting, views, commands and conics.

UNIT-III: **12Hrs**

Projections of Regular Solids (Conventional and Computer Aided)

Sections or Sectional views of Right Regular Solids, Prism, Cylinder, Pyramid, Cone, and Computer aided projections of solids, sectional views

UNIT-IV: **10Hrs**

Development of Surfaces (Conventional)

Prism, Cylinder, Pyramid and Cone.

UNIT-V: **13Hrs**

Isometric Projections (Conventional and Computer Aided)

Principles of Isometric Projection, Isometric Scale, Isometric Views, Conventions, Isometric Views of Lines, Plane Figures, Simple and Compound Solids, Isometric Projection of objects having non, isometric lines. Conversion of Isometric Views to Orthographic Views and Vice-versa, Conventions. Conversion of orthographic projection in to isometric view.

Note:

1. The End Semester Examination will be in conventional mode.
2. CIE–I will be in conventional mode.
3. CIE–II will be using Computer.

Textbooks:

1. Engineering Drawing, N.D.Bhatt, Charotar,54th Edition, 2023.
2. Engineering Drawing and graphics Using AutoCAD, T. Jeyapoovan and Vikas, S. Chand and company Ltd., 3rdEdition,2010.

References:

1. Engineering Drawing, Basant Agrawal and C.M.Agrawal, McGraw Hill,3rdEdition,2019.
2. Engineering Graphics and Design, WILEY, John Wiley and Sons Inc,3rdEdition, 2020.
3. Engineering Drawing, M.B.Shah and B.C.Rane,Pearson,2ndEdition, 2009.
4. Engineering Drawing, N.S.Parthasarathy and Vela Murali, Oxford,1st Edition,2015.
5. Computer Aided Engineering Drawing, K.Balaveera Reddy, CBS Publishers, 2nd Edition,2015.

Course Outcomes: After completion of the course, the student should be able to

CO-1. Understand and apply the principles of orthographic and isometric projections.
CO-2. Create sectional views and dimensioned drawings using BIS standards.
CO-3. Use CAD software to generate 2D engineering drawings.
CO-4. Visualize and construct solid models from 2D views.
CO-5. Demonstrate drafting skills for practical and industrial applications.

Course Articulation Matrix:

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**1**	**2**
CO-1	3	3	3	2	2	-	-	-	1	-	1	1	2	-
CO-2	3	3	3	2	2	-	-	-	1	2	2	1	2	-
CO-3	1	1	3	3	3	-	-	-	1	1	3	2	-	3
CO-4	2	3	3	3	3	-	-	-	2	1	3	2	-	2
CO-5	1	2	2	3	3	-	2	2	2	3	3	2	-	1

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

I B.Tech. I Semester

Course Name	:	**ELECTRICAL CIRCUITS – I**				
Course Code	:	**2502101**	L	T	P	C
Course Category	:	**Theory**	3	0	0	3

CIE: 40 SEE: 60 Total: 100

Course Objectives:

- To gain knowledge in circuits and to understand the fundamentals of derived circuit laws.
- Develop the ability to analyze single-phase AC circuits
- To understand the concepts of Network Theorems.
- To learn steady state and transient analysis of 3-phasecircuits.
- To understand the concepts of coupled circuits and networks.

UNIT-I: **12 Hrs**

Network Elements & Laws: Active elements, Independent and dependent sources, Ohms law, Passive elements - R, L and C, Energy stored in inductance and capacitance, Kirchhoff's laws, Source transformations, Star-delta transformations, Node voltage method and Mesh current method**.**

UNIT-II: **12 Hrs**

Single-Phase Circuits: RMS and average values of periodic sinusoidal and non- sinusoidal waveforms, Phasor representation, Steady-state response of series, parallel and series-parallel circuits. Impedance, Admittance, Resonance: Series and parallel circuits, Bandwidth and Q-factor, Applications.

UNIT-III: **13 Hrs**

Network theorems: Superposition theorem, Thevenin's theorem, Norton's theorems (AC & DC), Maximum power transfer theorem, Tellegen's theorem, Compensation theorem, Millman's theorem and Reciprocity theorem (DC only).

UNIT-IV: **11Hrs**

Poly-phase Circuits: Analysis of balanced and unbalanced 3-phase circuits, Star and delta connections, Measurement of three-phase power for balanced and unbalanced loads.

UNIT-V: **12 Hrs**

Coupled circuits: Concept of self and mutual inductance, Dot convention, Coefficient of coupling, Analysis of circuits with mutual inductance.

Topological Description of Networks: Graph, tree, chord, cut-set, incident matrix, circuit matrix and cut-set matrix.

Textbooks:

1. Van Valkenburg M.E, "Network Analysis", Prentice Hall of India, 3rd Edition, 2019.
2. Ravish R Singh, "Network Analysis and Synthesis", McGraw Hill, 2nd Edition, 2019.

References:

1. B.Subramanyam, “Electric Circuit Analysis”, Dreamtech Press & Wiley, 2021.
2. James W.Nilsson, Susan A. Riedel,“ElectricCircuits”,Pearson,11th Edition,2020.
3. A Sudhakar, Shyammohan S Palli, “Circuits and Networks: Analysis and Synthesis”, McGraw Hill, 5th Edition, 2017.
4. Jagan N.C, Lakshrninarayana C., “Network Analysis”, B.S. Publications, 3rd Edition, 2014.
5. William Hayt H, Kimmerly Jack E. And Steven Durbin M, “Engineering Circuit Analysis”, McGraw Hill, 6th Edition, 2002.
6. Chakravarthy A., “Circuit Theory”, Dhanpat Rai & Co., First Edition, 1999.

Course Outcomes: After completion of the course, the student should be able to

CO-1. Understand network analysis, techniques using mesh and node analysis.
CO-2. Evaluate steady state and transient behavior of single phase circuits.
CO-3. Analyze electric circuits using network theorems.
CO-4. Evaluate steady state and transient behavior of three phase circuits.
CO-5. Analyze electric circuits using network theorems and concepts of coupled circuits.

Course Articulation Matrix:

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**1**	**2**
CO-1	1	2	3	3	3	2	2	-	-	-	2	2	2	2
CO-2	2	1	3	2	3	1	3	-	-	-	3	2	3	3
CO-3	1	2	3	3	3	2	3	-	-	-	2	2	3	3
CO-4	2	2	3	2	3	2	-	-	-	-	2	3	3	3
CO-5	3	2	2	2	3	2	2	-	-	-	2	2	3	3

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

I B.Tech. I Semester

Course Name	:	**ENGINEERING PHYSICS LABORATORY**				
Course Code	:	**2599107**	L	T	P	C
Course Category	:	**Laboratory**	0	0	2	1

CIE: 40 SEE: 60 Total: 100

Course Objectives:

- To provide hands-on experience in studying the electrical, optical, and magnetic properties of various materials.
- To understand the working principles and characteristics of semiconductor devices like PN junction diodes, LEDs, and solar cells.
- To explore advanced concepts in modern physics through experiments on lasers, photoelectric effect, and optical fibers.
- To develop experimental skills for measuring fundamental constants and material parameters.
- To improve abilities in scientific data collection, result analysis, and technical reporting.

List of experiments:

1. Determination of Energy gap of a semiconductor.
2. I-V Characteristics of P-N junction diode.
3. B-H curve of a ferromagnetic material.
4. (a) V-I Characteristics of LED diode.
   (b) V-I Characteristics of a solar cell.
5. Determination of wavelength of laser using diffraction grating.
6. Determination of Planck's constant using photo electric effect.
7. Determination of magnetic moment of a bar magnet and horizontal earth magnetic field.
8. (a) Determine the Numerical Aperture of an Optical Fiber
   (b) Determine the bending losses of an Optical Fiber
9. Determination of dielectric constant of a given material.
10. Determination of Hall coefficient and carrier concentration of a given semiconductor.

Note: Any 8 experiments are to be performed.

References:

1. Dr. Y. Aparna & Dr. K. Venkateswara Rao, Laboratory Manual of Engineering Physics, V.G.S Publishers
2. G.L. Squires – *Practical Physics*, Cambridge University Press, 2012.
3. S. Balasubramanian, M.N. Srinivasan "A Text book of Practical Physics"- S Chand Publishers,2017

Course Outcomes:

After completing this laboratory course, the student will be able to:

CO1: Determine key semiconductor parameters such as energy gap, Hall coefficient, and carrier concentration.

CO2: Analyze the I–V characteristics of PN junction diodes, LEDs, and solar cells for different applications.

CO3: Determine key electrical, magnetic, and optical properties of semiconductors and other functional materials.

CO4: Demonstrate the working principles of lasers, optical fibers, and the photoelectric effect through precise measurements.

CO5: Apply scientific methods for accurate experimentation, data analysis, and preparation of technical reports.

Course Articulation Matrix:

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**1**	**2**
CO-1	2	3	2	1		-	-	-	-	-	-	2	2	2
CO-2	2	1	1	-	-	-	-	-	-	-	-	2	2	1
CO-3	3	2	1	-	--	-	-	-	-	-	-	2	-	-
CO-4	2	2	1	-	-	-	-	-	-	-	-	2	-	1
CO-5	2	2	2	1	-	-	-	-	-	-	-	-	1	1

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

I B.Tech. I Semester

Course Name	:	**C PROGRAMMING AND DATA STRUCTURES LABORATORY**				
Course Code	:	**2505104**	L	T	P	C
Course Category	:	**Laboratory**	0	0	2	1

CIE: 40 SEE: 60 Total: 100

Course Objectives:

- To apply the fundamentals of C programming to solve simple mathematical and logical problems.
- To develop programs using functions, recursion, arrays, and structures.
- To implement searching and sorting algorithms for computational problem-solving.
- To design and implement linked lists and perform insertion, deletion, and traversal operations.
- To implement stacks and queues using arrays and pointers for problem-solving.

Week-1 & 2

1) Write a C program to find the roots of a quadratic equation.
2) Write a C program, which takes two integer operands and one operator from the user, performs the operation and then prints the result. (Consider the operators +,-,*, /, % and use Switch Statement)
3) Write a C program to find the sum of the individual digits of a positive integer.

Week-3 & 4

4) Write a C program to find the factorial of a given integer.
5) Write a C program to find the GCD (greatest common divisor) of two given integers.
6) Write a C program to find both the largest and smallest number in a list of integers.
7) Write a C program that uses functions to perform the following:
i) Addition of Two Matrices ii) Multiplication of Two Matrices

Week-5 & 6

8) Write a C program to construct a pyramid of numbers.
9) Write a C program that uses functions to perform the following operations:
i) Reading a complex number ii) Writing a complex number
iii) Addition of two complex numbers
iv) Multiplication of two complex numbers (Note: represent complex number using a structure.)

Week- 7 & 8

10) Write a C program that implements the following sorting methods to sort a given list of integers in ascending order i) Bubble sort ii) Selection sort iii)Insertion sort
11) Write C programs that use both recursive and non-recursive functions to perform the following searching operations for a Key value in a given list of integers: i) Linear search ii) Binary search

Week- 9 & 10

12. Write a C program that uses functions to perform the following operations on singly linked list.
i) Creation ii) Insertion iii) Deletion iv) Traversal

Week-11 & 12

13. Write C programs that implement a stack (its operations) using i) Arrays ii) Pointers
14. Write C programs that implement Queue (its operations) using i) Arrays ii) Pointers

Textbooks:

1. C Programming & Data Structures, B.A. Forouzan and R. F. Gilberg, Third Edition, Cengage Learning.
2. Let us C, Yeswanth Kanitkar
3. C Programming, Balaguruswamy.

Reference Books:

1. Brian W. Kernighan and Dennis M. Ritchie, The C Programming Language, Prentice Hall of India
2. E. Balagurusamy, Computer Fundamentals and C, 2nd Edition, McGraw-Hill

Course Outcomes:

After completion of the Course, the student should be able to

CO-1. Apply C programming constructs (operators, I/O, control) to solve mathematical problems.
CO-2. Develop solutions using functions, recursion, arrays, and structures.
CO-3. Implement searching and sorting algorithms using iterative/recursive methods.
CO-4. Design and implement linked list operations (creation, insertion, deletion, traversal).
CO-5. Create stack and queue applications using arrays and pointers.

Course Articulation Matrix:

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**1**	**2**
CO-1	3	2	1	-	2	-	-	-	1	-	-	-	3	1
CO-2	3	3	2	-	3	-	-	-	1	-	-	-	3	2
CO-3	3	3	3	-	3	1	-	-	2	-	-	-	3	3
CO-4	2	3	3	-	3	1	-	-	2	-	-	-	3	3
CO-5	2	3	3	1	3	2	-	-	2	-	-	-	3	3

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

I B.Tech. I Semester

Course Name	:	**ELECTRICAL CIRCUITS – I LABORATORY**				
Course Code	:	**2502102**	L	T	P	C
Course Category	:	**Laboratory**	0	0	2	1

CIE: 40 SEE: 60 Total: 100

Course Objectives:

- To verify the Kirchhoff's laws.
- To measure the electrical parameters for different types of DC circuits
- To measure the electrical parameters for different types AC circuits
- To study the phenomenon of series and parallel resonance.
- To measure three phase Active and Reactive power.

The following experiments are required to be conducted as compulsory

1. Verification of KVL and KCL.
2. Verification of Thevenin's and Norton's theorem.
3. Verification of Superposition and Reciprocity theorem.
4. Calculations and Verification of Impedance and Current of RL, RC and RLC series circuits.
5. Verification of Series Resonance.
6. Determination of Co-efficient of Coupling and Separation of Self and Mutual inductance in a Coupled Circuits.
7. Verification of Maximum Power Transfer Theorem
8. Measurement of Active Power for Star and Delta connected balanced loads.

Any two experiments from the given list

1. Verification of Millman's Theorem.
2. Verification of Compensation theorem.
3. Measurement of Reactive Power for Star and Delta connected balanced loads.
4. Verification of Parallel Resonance

Course Outcomes: After completion of the course, the student should be able to

CO-1.Verify the Kirchhoff's laws.
CO-2.Verify the basic Electrical circuit's parameters with DC excitation.
CO-3.Verify the basic Electrical circuit's parameters with AC excitation.
CO-4.Analyze the phenomenon of series and parallel resonance.
CO-5.Measure active and reactive power in star and delta connected balanced loads.

Course Articulation Matrix:

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**1**	**2**
CO-1	2	1	2	2	2	-	-	1	-	1	2	3	2	-
CO-2	2	1	2	2	2	-	-	1	-	1	2	2	-	2
CO-3	2	1	1	2	2	-	-	1	-	1	2	3	1	-
CO-4	2	1	2	1	2	-	-	1	-	1	1	3	-	1
CO-5	2	1	2	2	2	-	-	1	-	1	2	3	-	2

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

I B.Tech. I Semester

Course Name	:	**IT WORKSHOP**				
Course Code	:	**2505105**	L	T	P	C
Course Category	:	**Laboratory**	0	0	2	1

CIE: 40 SEE: 60 Total: 100

Course Objectives:

- Training on PC Hardware, troubleshooting
- Training Internet & World Wide Web and
- Training Productivity tools including Word, Excel, PowerPoint and Publisher.
- Training on Safeguard computer systems from viruses/worms
- Training on Document/ Presentation preparation

PC Hardware

Task 1: Identify the peripherals of a computer, components in a CPU and its functions. Draw the block diagram of the CPU along with the configuration of each peripheral and submit to your instructor.

Task 2: Every student should disassemble and assemble the PC back to working condition. Lab instructors should verify the work and follow it up with a Viva. Additionally, students should watch the video, which demonstrates the process of assembling a PC. A video would be given as part of the course content.

Task 3: Every student should individually install Microsoft Windows on their personal computer. The lab instructor should verify the installation and follow it up with a Viva.

Task 4: Every student should install Linux on the computer. This computer should have Windows installed. The system should be configured as a dual-boot with both Windows and Linux. Lab instructors should verify the installation and follow it up with a Viva

Internet & World Wide Web

Task1: Orientation & Connectivity Boot Camp: Students should get connected to their Local Area Network and access the Internet. In the process, they configure the TCP/IP settings. Finally, students should demonstrate to the instructor how to access the websites and email. If there is no internet connectivity, preparations need to be made by the instructors to simulate the WWW on the LAN.

Task 2: Web Browsers, Surfing the Web: Students customize their web browsers with the LAN proxy settings, bookmarks, search toolbars and pop up blockers. Also, plug-ins like Macromedia Flash and JRE for applets should be configured.

Task 3: Search Engines & Netiquette: Students should know what search engines are and how to use them. A few topics would be assigned to the students, for which they would need to conduct a Google search. This should be demonstrated to the instructors by the student.

Task 4: Cyber Hygiene: Students would be exposed to the various threats on the internet and would be asked to configure their computers to be safe on the internet. They need to customize their browsers to block pop-ups, block ActiveX downloads to avoid viruses and/or worms.

LaTeX and WORD

Task 1 – Word Orientation: The mentor needs to give an overview of LaTeX and Microsoft (MS) office or equivalent (FOSS) tool word: Importance of LaTeX and MS office or equivalent (FOSS) tool Word as word Processors, Details of the four tasks and features that would be covered in each, Using LaTeX and word – Accessing, overview of toolbars, saving files, Using help and resources, rulers, format painter in word.

Task 2: Using LaTeX and Word to create a project certificate. Features to be covered:- Formatting Fonts in Word, Drop Cap in Word, Applying Text effects, Using Character Spacing, Borders and Colours, Inserting Header and Footer, Using Date and Time option in both LaTeX and Word.

Task 3: Creating project abstract Features to be covered:-Formatting Styles, Inserting table, Bullets and Numbering, Changing Text Direction, Cell alignment, Footnote, Hyperlink, Symbols, Spell Check, Track Changes.

Task 4: Creating a Newsletter: Features to be covered include Table of Contents, Newspaper columns, Images from files and clip art, the Drawing toolbar and Word Art, Formatting Images, Text Boxes, Paragraphs, and Mail Merge in Word.

Excel

Excel Orientation: The mentor needs to emphasize the importance of MS Office or an equivalent (FOSS) tool, such as Excel, as a Spreadsheet tool. They should provide details of the four tasks and features that will be covered in each. Using Excel – Accessing, overview of toolbars, saving Excel files, using help and resources.

Task 1: Creating a Scheduler - Features to be covered: Gridlines, Format Cells, Summation, auto fill, Formatting Text.

Task 2: Calculating GPA - Features to be covered:- Cell Referencing, Formulae in Excel – average, std. Deviation, Charts, Renaming and Inserting worksheets, Hyperlinking, Count function, LOOKUP/VLOOKUP.

Task 3: Split cells, freeze panes, group and outline, Sorting, Boolean and logical operators, and Conditional formatting.

PowerPoint

Task 1: Students will be working on basic PowerPoint utilities and tools, which help them create basic PowerPoint presentations. PPT Orientation, Slide Layouts, Inserting Text, Word Art, Formatting Text, Bullets and Numbering, Auto Shapes, Lines and Arrows in PowerPoint.

Task 2: Interactive Presentations - Hyperlinks, Inserting Images, Clip Art, Audio, Video, Objects, Tables, and Charts.

Task 3: Master Layouts (slide, template, and notes), Types of views (basic, presentation, slide slotter, notes etc), and Inserting – Background, textures, Design Templates, Hidden slides.

Reference Books:

1. Comdex Information Technology course tool kit Vikas Gupta, WILEY Dreamtech
2. The Complete Computer upgrade and repair book, 3rd edition Cheryl A Schmidt, WILEY Dreamtech
3. Introduction to Information Technology, ITL Education Solutions limited, Pearson Education.
4. PC Hardware - A Handbook – Kate J. Chase PHI (Microsoft)
5. LaTeX Companion – Leslie Lamport, PHI/Pearson.
6. IT Essentials PC Hardware and Software Companion Guide Third Edition by David Anfinson and Ken Quamme. – CISCO Press, Pearson Education.
7. IT Essentials PC Hardware and Software Labs and Study Guide Third Edition by Patrick Regan – CISCO Press, *Pearson Education.*

Course Outcomes:

After completion of the Course, the student should be able to

CO-1. Perform Hardware troubleshooting
CO-2. Understand Hardware components and inter dependencies
CO-3. Safeguard computer systems from viruses/worms
CO-4. Document/ Presentation preparation
CO-5. Perform calculations using spreadsheets

Course Articulation Matrix:

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**1**	**2**
CO-1	3	2	2	1	2	-	-	-	-	-	-	1	2	1
CO-2	3	2	2	2	3	1	-	-	-	1	-	1	3	2
CO-3	2	2	3	2	2	1	-	-	-	1	-	2	3	2
CO-4	2	1	2	2	3	-	-	-	-	2	-	2	2	3
CO-5	2	1	2	2	3	1	-	-	-	2	-	3	2	3

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

I Year II Semester

I B.Tech. II Semester

Course Name	:	**ORDINARY DIFFERENTIAL EQUATIONS AND VECTOR CALCULUS**				
Course Code	:	**2599201**	L	T	P	C
Course Category	:	**Theory**	3	0	0	3

CIE: 40 SEE: 60 Total: 100

Course Objectives:

To learn

- Methods of solving the differential equations of first order and applications.
- Methods of solving the differential equations of higher order
- Concept, properties of Laplace transforms, solving ordinary differential equations using Laplace transforms techniques
- The physical quantities involved in engineering field related to vector valued functions
- The basic properties of vector valued functions and their applications to line, surface and volume integrals

UNIT-I: First Order Ordinary Differential Equations **8 Hrs**

Exact differential equations – Equations form: Homogeneous form and $yf(xy)dx + xg(xy)dy = 0$ reducible to exact differential equations – linear and Bernoulli's equations. Applications: Newton's law of cooling – Law of natural growth and decay.

UNIT-II: Ordinary Differential Equations of Higher Order **10 Hrs**

Higher order linear differential equations with constant coefficients: Non-Homogeneous terms of the type V(x): e^x, sin ax , cos ax, polynomials in x, $e^{ax}V(x)$ and $xV(x)$ – Method of variation of parameters.

UNIT-III: Laplace Transforms **10 Hrs**

Laplace Transforms: Laplace Transform of standard functions – First shifting theorem – Second shifting theorem-Laplace transforms of functions multiplied by $'t'$ and divided by $'t'$ – Evaluation of integrals by Laplace transforms – Inverse Laplace transform by different methods, convolution theorem (without proof).

Applications: solving Initial value problems by Laplace Transform method**.**

UNIT-IV: Vector Differentiation **10 Hrs**

Vector point functions and scalar point functions – Gradient – Divergence and Curl – Directional derivatives – Angle between two surfaces – Scalar potential functions – Solenoidal and Irrotational vectors.

UNIT-V: Vector Integration **10 Hrs**

Line, Surface and Volume Integrals. Theorems of Green, Gauss and Stokes (without proofs) and their applications.

Text Books:

1. B.S. Grewal, Higher Engineering Mathematics, Khanna Publishers, 36th Edition, 2010.
2. R.K. Jain and S.R.K. Iyengar, Advanced Engineering Mathematics, Narosa Publications, 5th Edition, 2016.

Reference Books:

1. Erwin Kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006.
2. G.B. Thomas and R.L. Finney, Calculus and Analytic geometry, 9thEdition, Pearson, Reprint,2002
3. N.P. Bali and Manish Goyal, A text book of Engineering Mathematics, Laxmi Publications, Reprint,2008.
4. H. K. Dass and Er. Rajnish Verma, Higher Engineering Mathematics, S Chand and Company Limited, New Delhi.

Course Outcomes: After learning the contents of this paper, the student must be able to

CO-1. Identify whether the given differential equation of first order is exact or not

CO-2. Solve higher differential equation and apply the concept of differential equation to real world problems.

CO-3. Use the Laplace Transforms techniques for solving Ordinary Differential Equations.

CO-4. Analyze and solve problems involving gradient, divergence, and curl in various fields such as physics and engineering

CO-5. Evaluate the Line, Surface and Volume integrals and converting them one to another

Course Articulation Matrix:

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**1**	**2**
CO-1	3	2	3	2	3	2	2	2	-	-	-	2	2	2
CO-2	3	2	3	2	3	2	2	2	-	-	-	2	2	3
CO-3	3	2	3	2	3	2	2	2	-	-	-	2	2	3
CO-4	3	2	3	2	3	2	2	2	-	-	-	2	3	3
CO-5	3	2	3	2	3	2	2	2	-	-	-	2	3	3

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

I B.Tech. II Semester

Course Name	:	**ENGINEERING CHEMISTRY**				
Course Code	:	**2599204**	L	T	P	C
Course Category	:	**Theory**	3	0	0	3

CIE: 40 SEE: 60 Total: 100

Course Objectives:

- To develop adaptability to new advances in Engineering Chemistry and acquire the essential skills to become a competent engineering professional.
- To understand the industrial significance of water treatment, fundamental principles of battery chemistry, and the impact of corrosion along with its control methods for structural protection.
- To impart foundational knowledge of various energy sources and their practical applications in engineering.
- To equip students with an understanding of smart materials, biosensors, and analytical techniques applicable in engineering, industrial, environmental, and biomedical fields.
- To acquire required knowledge about engineering materials like cement, UV-Visible, IR spectroscopy

UNIT-I: Water and its treatment: 12 Hrs

Introduction- Hardness, types, degree of hardness and units. Estimation of hardness of water by complexometric method - Numerical problems for Temporary and permanent hardness . Potable water and its specifications - Steps involved in the treatment of potable water - Disinfection of potable water by chlorination and break-point chlorination. Defluoridation- Nalgonda technique.

Boiler troubles: Scales, Sludges and Caustic embrittlement. Internal treatment of boiler feed water - Calgon conditioning, Phosphate conditioning, Colloidal conditioning. External treatment methods - Softening of water by ion- exchange processes. Desalination of brackish water – Reverse osmosis.

UNIT-II: Electrochemistry and Corrosion: 13Hrs

Introduction- Electrode potential, standard electrode potential, Nernst equation (no derivation), electrochemical cell - Galvanic cell, cell representation, EMF of cell - Numerical problems. Types of electrodes, reference electrodes - Primary reference electrode - Standard Hydrogen Electrode (SHE), Secondary reference electrode - Calomel electrode. Construction, working and determination of pH of unknown solution using SHE and Calomel electrode.

Corrosion: Introduction- Definition, causes and effects of corrosion – Theories of corrosion, chemical and electrochemical theories of corrosion, Types of corrosion: galvanic, water-line and pitting corrosion. Factors affecting rate of corrosion - Nature of the metal (Position of metal in galvanic series, Purity of metal, Nature of corrosion product), Nature of the corroding environment (Temperature, Humidity, and pH).. Corrosion control methods - Cathodic protection Methods - Sacrificial anode and impressed current methods.

UNIT–III: Energy Sources: 13Hrs

Batteries: Introduction – Classification of batteries - Primary, secondary and reserve batteries with examples. Construction, working and applications of Zn-air and Lithium ion battery. Fuel Cells – Differences between a battery and a fuel cell, Construction and applications of Direct Methanol Fuel Cell (DMFC).

Fuels: Introduction and characteristics of a good fuel, Calorific value – Units- HCV, LCV- Dulongsformula- Numerical problems.

Fossil fuels: Introduction, Classification, Petroleum- Refining of Crude oil, Cracking - Types of cracking - Moving bed catalytic cracking. LPG and CNG composition and uses.

Synthetic Fuels:Fischer-Tropsch process, Introduction and applications of Hythane and Green Hydrogen.

UNIT - IV: Polymers: 13Hrs

Definition - Classification of polymers: Based on origin and tacticity with examples – Types of polymerization - Addition and condensation polymerization.

Plastics, Elastomers and Fibers: Definition and applications (PVC, Buna-S, Nylon-6,6),preparation ,properties and applications of Bakelite. Differences between themoplastics and thermo setting plastics,

Conducting polymers: Definition and Classification with examples - Mechanism of conduction in trans poly-acetylene and applications of conducting polymers.
Biodegradable polymers: Polylactic acid and its applications.

UNIT-V- Advanced Functional Materials: **10Hrs**

Cement: Portland cement, its composition, setting and hardening.
Biosensor - Definition, Amperometric Glucose monitor sensor. Interpretative spectroscopic applications of UV-Visible spectroscopy for Analysis of pollutants in dye industry, IR spectroscopy in night vision-security, Pollution Under Control-CO sensor (Passive Infrared detection).

Text Books:

1. Engineering Chemistry by P.C. Jain and M. Jain, Dhanpatrai Publishing Company, 2010.
2. Engineering Chemistry by Rama Devi,Dr.P.Aparna and Rath, Cengage learning, 2025.

Reference Books:

1. Engineering Chemistry: by Thirumala Chary Laxminarayana& Shashikala, Pearson Publications (2020)
2. Engineering Chemistry by Shashi Chawla, Dhanpatrai and Company (P) Ltd. Delhi 2011.
3. Engineering Chemistry by Shikha Agarwal, Cambridge University Press, Delhi 2015.
4. Engineering Analysis of Smart Material Systems by Donald J. Leo, Wiley, 2007.
5. Challenges and Opportunities in Green Hydrogen by Editors: Paramvir Singh, Avinash Kumar Agarwal, Anupma Thakur, R.K Sinha.
6. Raman Spectroscopy in Human Health and Biomedicine, https://www.worldscientific.com/doi/epdf/10.1142/13094
7. E-Content- https://doi.org/10.1142/13094 | October 2023
8. E-books: https://archive.org/details/EngineeringChemistryByShashiChawla/page/n11/mode/2u

Course Outcomes: After learning the contents of this paper, the student must be able to

CO-1. Students will be able to understand the fundamental properties of water and its applications in both domestic and industrial purposes.
CO-2. Students will gain basic knowledge of electrochemical processes and their relevance to corrosion and its control methods.
CO-3. Students will comprehend the significance and practical applications of batteries and various energy sources, enhancing their potential as future engineers and entrepreneurs
CO-4. Students will learn the basic concepts and properties of polymers and other engineering materials
CO-5. Students will be able to apply the principles of UV-Visible, IR spectroscopy in analyzing pollutants in dye industries and biomedical applications

Course Articulation Matrix:

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO-1	3	2	2	-	2	2	-	-	2	-	2	2	3	2
CO-2	3	2	3	2	3	-	-	-	2	2	2	2	3	3
CO-3	3	2	3	2	3	-	2	2	3	2	3	3	3	3
CO-4	2	2	2	-	2	-	2	-	2	-	2	2	2	2
CO-5	2	2	3	2	3	-	-	-	3	2	3	2	2	3

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

I B.Tech. II Semester

Course Name	:	**ENGLISH FOR SKILL ENHANCEMENT**				
Course Code	:	**2599205**	L	T	P	C
Course Category	:	**Theory**	3	0	0	3

CIE: 40 SEE: 60 Total: 100

Course Objectives:

- Improve their vocabulary.
- Use appropriate sentence structures in their oral and written communication.
- Develop their reading and study skills.
- Equip students to write paragraphs, essays, précis and draft letters.
- Acquire skills for technical report writing

UNIT-I: **12 Hrs**

Theme: **Perspectives**
Lesson on 'The Generation Gap' by Benjamin M. Spock from the prescribed textbook titled *English for the Young in the Digital World* published by Orient Black Swan Pvt. Ltd.

Vocabulary: The Concept of Word Formation -The Use of Prefixes and Suffixes - Words Often Misspelt - Synonyms and Antonyms

Grammar: Identifying Common Errors in Writing with Reference to Parts of Speech particularly Articles and Prepositions – Degrees of Comparison

Reading: Reading and Its Importance- Sub Skills of Reading – Skimming and Scanning.

Writing: Sentence Structures and Types -Use of Phrases and Clauses in Sentences- Importance of Proper Punctuation- Techniques for Writing Precisely –Nature and Style of Formal Writing.

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UNIT-II **12Hrs**

Theme: **Digital Transformation**
Lesson on '*Emerging Technologies*' from the prescribed textbook titled *English for the Young in the Digital World* published by Orient BlackSwan Pvt. Ltd.

Vocabulary: Homophones, Homonyms and Homographs

Grammar: Identifying Common Errors in Writing with Reference to Noun-pronoun Agreement and Subject-verb Agreement.

Reading: Reading Strategies-Guessing Meaning from Context – Identifying Main Ideas – Exercises for Practice

Writing: Paragraph Writing – Types, Structures and Features of a Paragraph - Creating Coherence – Linkers and Connectives - Organizing Principles in a Paragraph – Defining- Describing People, Objects, Places and Events – Classifying- Providing Examples or Evidence - Essay Writing - Writing Introduction and Conclusion.

UNIT–III: **12Hrs**

Theme: **Attitude and Gratitude**
Poems on '*Leisure'* by William Henry Davies and '*Be Thankful'* - Unknown Author from the prescribed textbook titled *English for the Young in the Digital World* published by Orient Black Swan Pvt. Ltd.

Vocabulary: Words Often Confused - Words from Foreign Languages and their Use in English.

Grammar: Identifying Common Errors in Writing with Reference to Misplaced Modifiers and Tenses.

Reading: Sub-Skills of Reading – Identifying Topic Sentence and Providing Supporting Ideas - Exercises for Practice.

Writing: Format of a Formal Letter-Writing Formal Letters E.g.., Letter of Complaint, Letter of Requisition, Job Application with CV/Resume –Difference between Writing a Letter and an Email - Email Etiquette.

UNIT - IV **13Hrs**

Theme: **Entrepreneurship**
Lesson on '*Why a Start-Up Needs to Find its Customers First*' by Pranav Jain from the prescribed textbook titled *English for the Young in the Digital World* published by Orient Black Swan Pvt. Ltd.

Vocabulary: Standard Abbreviations in English – Inferring Meanings of Words through Context – Phrasal Verbs – Idioms.

Grammar: Redundancies and Clichés in Written Communication – Converting Passive to Active Voice and Vice-Versa.

Reading: Prompt Engineering Techniques– Comprehending and Generating Appropriate Prompts - Exercises for Practice

Writing: Writing Practices- Note Making-Précis Writing.

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UNIT-V- **10Hrs**

Theme: **Integrity and Professionalism**
Lesson on '*Professional Ethics*' from the prescribed textbook titled *English for the Young in the Digital World* published by Orient BlackSwanPvt. Ltd.

Vocabulary: Technical Vocabulary and their Usage– One Word Substitutes – Collocations.

Grammar: Direct and Indirect Speech **-** Common Errors in English (Covering all the other aspects of grammar which were not covered in the previous units)

Reading: Survey, Question, Read, Recite and Review (SQ3R Method) – Inferring the Meaning and Evaluating a Text- Exercises for Practice

Writing: ***Report Writing - Technical Reports- Introduction – Characteristics of a Report – Categories of Reports Formats- Structure of Reports (Manuscript Format) - Types of Reports - Writing a Technical Report.***

Text Books:

1. Board of Editors. *2025. English for the Young in the Digital World.* Orient Black Swan Pvt. Ltd.

Reference Books:

1. Swan, Michael. (2016). *Practical English Usage.* Oxford University Press. New Edition.
2. *Karal, Rajeevan. 2023. English Grammar Just for You.* Oxford University Press. New Delhi
3. 2024*. Empowering with Language: Communicative English for Undergraduates.* Cengage Learning India Pvt. Ltd. New Delhi
4. Sanjay Kumar &Pushp Lata. 2022. *Communication Skills – A Workbook.* Oxford University Press. New Delhi
5. Wood,F.T. (2007). *Remedial English Grammar.* Macmillan.
6. Vishwamohan, Aysha. (2013). *English for Technical Communication for Engineering Students.* Mc Graw-Hill Education India Pvt. Ltd.

Course Outcomes: After learning the contents of this paper, the student must be able to

CO1: Choose appropriate vocabulary in their oral and written communication.
CO2: Demonstrate their understanding of the rules of functional grammar and sentence structures.
CO3: Develop comprehension skills from known and unknown passages.
CO4: Write paragraphs, essays, précis and draft letters.
CO5: Write abstracts and reports in various contexts.

Course Articulation Matrix:

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**1**	**2**
CO-1	-	-	-	-	-	-	1	2	2	2	3	-	-	2
CO-2	-	-	-	-	-	-	2	2	3	2	2	-	-	2
CO-3	-	-	-	-	-	-	2	2	2	1	3	-	-	1
CO-4	-	-	-	-	-	-	2	3	2	3	2	-	-	2
CO-5	-	-	-	-	-	-	3	2	3	3	3	-	-	2

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

I B.Tech. II Semester

Course Name	:	**PYTHON PROGRAMMING**				
Course Code	:	**2505201**	L	T	P	C
Course Category	:	**Theory**	3	0	0	3

CIE: 40 SEE: 60 Total: 100

Course Objectives:

- To introduce programming languages, program development steps, problem-solving approaches, algorithms, flowcharts, and the basics of Python.
- To understand Python data types, variables, operators, and input/output operations.
- To apply control statements and arrays for computational problem solving.
- To design modular programs using functions, recursion, and variable scope.
- To implement exception handling and file operations for reliable applications.

UNIT-I: 12 Hrs

Introduction To Programming Languages: Computer definition, Computer system, Programming languages: Types, Uses. Program development steps, Algorithm, Flowchart, Solving simple problems.
Introduction To Python: History of Python, Features of Python, Flavors of Python, Execution of a Python Program, Python Virtual Machine, Viewing the byte code, Comparisons between C and Python, Comments in Python, Docstrings

UNIT-II: 12 Hrs

Datatypes In Python: Built-in data types: None, Numeric and Boolean data type, Sequences in Python: strings, bytes, bytearrays, list, tuple, sets, frozenset, dictionaries. User-defined data types, constants in Python, identifiers and reserved words, naming conventions in Python.
Variables And Operators: Understanding Python variables, multiple variable declarations; Operators in Python - Arithmetic operators, Assignment operators, Relational Operators, Logical operators, Boolean Operators, Bitwise operators, Membership operators, Identity operators. Operator Precedence and Associativity.
Input Output Statements: Output statements, Input Statements and Command Line Arguments, A word on Indentation.
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UNIT-III: 12 Hrs

Control Statements: The if Statement, The if else statements, The if…elif…else statements, while loop, for loop, Infinite loop, Nested Loops, else suite, break, continue, pass statement, assert statement.
Arrays: Introduction to Arrays, Creating an Array, Types of arrays, Importing the array Module and NumPy, Indexing and Slicing on numpyArrays, Mathematical Operations on Arrays, Comparison of Arrays.

UNIT-IV: 12 Hrs

Functions: Introduction, Defining a Function, calling a function, Formal and Actual Arguments, Positional Arguments, keyword Arguments, Default Arguments, variable length arguments, local and global variables, global keyword, nesting of functions and Recursion.

UNIT-V: 12 Hrs

Exceptions In Python: Errors in Python Program - Compile-Time Errors, Runtime Errors, Logical Errors; Exceptions, Exception Handling, Types of Exceptions, and Keywords in Exception Handling.
Files In Python: Types of Files in Python, Opening a File, Closing a File, Random File accessing methods- seek() and tell().

Textbooks:

1. R. Nageswara Rao, “Core Python Programming”, Dreamtech
2. Fundamentals of Python First Programs, Kenneth A. Lambert, 2nd Edition, Cengage**.**

References:

1. Allen B. Downey, “Think Python: How to Think like a Computer Scientist”, 2nd edition, Updated for Python 3, Shroff/O‘Reilly Publishers, 2016.
2. Core Python Programming, W.Chun, Pearson.

Course Outcomes: After completion of the course, the student should be able to

CO-1. Understand problem-solving strategies, algorithms, flowcharts, and basics of Python programming.
CO-2. Apply Python data types, variables, operators, and I/O operations to solve simple problems.
CO-3. Use control structures and arrays to implement Python programs.
CO-4. Develop modular solutions using functions, recursion, and scope management.
CO-5. Implement exception handling and file operations for building reliable Python applications.

Course Articulation Matrix:

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**1**	**2**
CO-1	3	2	1	-	1	-	-	-	-	1	-	-	3	1
CO-2	3	3	2	-	2	-	-	-	-	1	-	-	3	2
CO-3	2	3	3	-	3	1	-	-	-	2	1	-	3	2
CO-4	2	2	2	-	2	1	-	-	-	2	2	-	2	2
CO-5	2	3	3	-	3	2	-	1		2	2	-	3	3

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

I B.Tech. II Semester

Course Name	:	**ELECTRICAL CIRCUITS – II**				
Course Code	:	**2502203**	L	T	P	C
Course Category	:	**Theory**	3	0	0	3

CIE: 40 SEE: 60 Total: 100

Course Objectives:

- To study the transient analysis of various R, L and C circuits for different inputs.
- To learn about two-port networks parameters and concept of filters.
- To understand the Laplace transformation of electrical circuits.
- To understand the concept of Fourier series and Integral.
- To learn about the concept of different filter

UNIT-I: 13 Hrs

Transient analysis: Transient response of R, L & C circuits, Formulation of integral differential equations, Initial conditions, Transient Response of RL, RC and RLC (series) networks for DC and sinusoidal excitations, Current locus diagrams of RL and RC series circuits with variation of various parameters.

UNIT-II: 12 Hrs

Two port network parameters: Open circuit impedance, short-circuit admittance, Transmission, Hybrid parameters & inter-relationships, Series, parallel and cascade connection of two port networks.

UNIT-III: 12 Hrs

Electrical circuit Analysis using Laplace Transforms: Application of Laplace Transforms to RL, RC and RLC (series) Networks for step, ramp and sinusoidal excitations.

UNIT-IV: 12 Hrs

Fourier Series and Integral: Fourier series representation of periodic functions, Symmetry conditions, Exponential Fourier series, Discrete spectrum, Fourier integral and its properties, Continuous spectrum, Application to simple networks.

UNIT-V: 11 Hrs

Filters: Classification of filters, T-Network, π Network, Characteristic of Filters, Constant-k Low Pass Filter, High-pass Filter, Band-pass Filter, Band-stop Filter.

Textbooks:

1. Van Valkenburg M.E, “Network Analysis”, Prentice Hall of India, 3rd Edition, 2019.
2. Ravish R Singh, “Network Analysis and Synthesis”, McGraw Hill, 2nd Edition, 2019.

References:

1. B.Subramanyam, “Electric Circuit Analysis”, Dreamtech Press & Wiley, 2021.
2. James W.Nilsson, Susan A. Riedel, “Electric Circuits”, Pearson,11th Edition, 2020.
3. A Sudhakar, Shyammohan S Palli, “Circuits and Networks: Analysis and Synthesis”, McGraw Hill, 5th Edition, 2017.
4. Jagan N.C, Lakshrninarayana C., “Network Analysis”, B.S.Publications, 3rd Edition, 2014.
5. William Hayt H, Kimmerly Jack E. and Steven Durbin M, “Engineering Circuit Analysis”, McGraw Hill, 6th Edition, 2002.
6. Chakravarthy A., “Circuit Theory”, Dhanpat Rai & Co., First Edition,1999.

Course Outcomes:

CO-1: Observe the response of various R, L and C circuits for different excitations.
CO-2: Examine the behavior of circuits using Laplace transforms and transfer function of single port network.
CO-3: Obtain two port network parameters and applications and design of various filters.
CO-4: Examine the behavior of circuits using Fourier transforms and integral.
CO-5: Classify and design various types of filters

Course Articulation Matrix:

CO	PROGRAM OUTCOMES(PO)												PROGRAMSPECIFIC OUTCOMES(PSO)	
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**1**	**2**
CO-1	3	3	2	-	2	-	2	-	-	-	2	-	3	2
CO-2	3	3	2	-	3	-	2	-	-	-	3	-	3	2
CO-3	3	2	3	-	3	-	2	-	-	-	3	2	3	3
CO-4	3	3	2	-	2	-	2	-	-	-	2	-	3	2
CO-5	3	2	3	-	3	-	2	-	-	-	3	2	3	3

(CorrelationofCourseOutcomeswithProgramOutcomesandProgramSpecificOutcomesusing ***mappinglevels1=Slight,2=Moderateand 3=Substantial****)*

I B.Tech. II Semester

Course Name	:	**ENGINEERING CHEMISTRY LABORATORY**				
Course Code	:	**2599206**	L	T	P	C
Course Category	:	**Laboratory**	0	0	2	1

CIE: 40 SEE: 60 Total: 100

Course Objectives:

- Students will understand and perform experiments based on core chemical principles relevant to engineering applications.
- Students will learn to estimate the hardness of water to assess its suitability for drinking purposes.
- Students will acquire the ability to perform acid-base titrations using instrumental methods such as conductometry, potentiometry, and pH metry.
- Students will gain hands-on experience in synthesizing polymers like Bakelite and Nylon – 6, 6 in the laboratory.
- Students will learn to determine the unknown concentration of potassium permanganate (KMnO4) using a calibration curve.

List of Experiments:

1. Estimation of Hardness of water by EDTA Complexometry method
2. Estimation of the concentration of strong acid by Conductometry.
3. Estimation of the concentration of strong and weak acid in an acid mixture by Conductometry.
4. Estimation of concentration of Fe^{+2}ion by Potentiometry using $KMnO_4$.
2. Estimation of concentration of strong acid with strong base by Potentiometry using quinhydrone
3. Determination of an acid concentration using pH meter.
4. Preparation of Bakelite.
5. 8.Preparation Nylon – 6, 6.
6. Corrosion: Determination of rate of corrosion of mild steel in the presence and absence of inhibitor.
7. Verification of Lambert-Beer's law using $KMnO_4$.
8. 11.Smart materials for Biomedical applications (Virtual lab)
9. Functioning of solar cell and its applications. (Virtual lab)

Reference Books:

1. Lab manual for Engineering chemistry by B. Ramadevi and P. Aparna, S Chand Publications, New Delhi (2022)
2. Vogel's text book of practical organic chemistry 5th edition
3. Inorganic Quantitative analysis by A.I. Vogel, ELBS Publications.
4. College Practical Chemistry by V.K. Ahluwalia, Narosa Publications Ltd. New Delhi (2007).

Course Outcomes:

CO-1. Students will develop practical skills through hands-on chemistry experiments relevant to engineering.
CO-2. Students will learn to determine important parameters such as water hardness and the corrosion rate of mild steel under various conditions.
CO-3. Students will be able to apply techniques like conductometry, potentiometry, and pH metryto determine concentrations or equivalence points in acid-base reactions.
CO-4. Students will gain experience in synthesizing polymers such as Bakelite.
CO-5. Students will understand the working principle of colorimetry and the relationship between absorbance and concentration (Beer-Lambert Law).

Course Articulation Matrix:

CO	PROGRAM OUTCOMES(PO)												PROGRAMSPECIFIC OUTCOMES(PSO)	
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**1**	**2**
CO-1	3	3	2	3	3	2	1	1	2	1	2	2	3	2
CO-2	3	3	3	2	3	2	2	2	2	2	3	2	3	3
CO-3	3	3	3	2	3	2	1	1	2	2	3	2	3	2
CO-4	3	2	2	1	2	1	1	2	2	1	2	2	2	2
CO-5	3	3	3	2	3	2	2	1	3	2	3	3	3	3

(Correlation of Course Outcomes with Program Outcomes and Program-Specific Outcomes using ***mappinglevels1=Slight,2=Moderate and 3=Substantial****)*

I B.Tech. II Semester

Course Name	:	**PYTHON PROGRAMMING LABORATORY**				
Course Code	:	**2505202**	L	T	P	C
Course Category	:	**Laboratory**	0	0	2	1

CIE: 40 SEE: 60 Total: 100

Course Objectives:

- To familiarize students with Python environment setup, syntax, operators, and basic input/output programs.
- To develop problem-solving skills using decision-making, control structures, and iterative constructs.
- To implement and manipulate Python data structures such as lists, tuples, dictionaries, and arrays (including NumPy).
- To design modular programs using user-defined functions and recursion.
- To implement file handling, exception handling, and command-line arguments for robust applications.

List of Experiments:

Week – 1:

1. Demonstrate the installation of Python for windows
2. Demonstrate the execution of Python program in Jupyter notebook and IDLE
3. Demonstrate the importing of NumPy, pandas, matplotlib and mention briefly about their purpose.
4. Write a Python program to assign a value to a variable and print the value of the variable.

Week – 2:

5. Write a Python program to assign different values to a single variable and print the value of the variable.
6. Write a Python program to assign multiple values to multiple variables and print the value of the variable.
7. Write a python program for printing values using format() method, % operator and format specifier.
8. Write a Python program to read two numbers from the user and print their sum and difference.

Week – 3:

9. Write a Python program implementing a simple calculator using all arithmetic operators.
10. Write a Python program illustrating the relation between the given two numbers using relational operators.
11. Write a Python program illustrating all logical operators.
12. Write a Python program illustrating all bitwise operators.

Week – 4:

13. Write a Python program to perform the swapping of the given two numbers:
    i) Using a temporary variable ii) Without using a temporary variable
14. Write a Python program to evaluate the following expressions:
    i) r // a + b * c – d / e ii) r //((a+b) * ((c- d) / e))
15. Write a Python program to calculate Simple Interest: SI = (P x T x R) /100

Week – 5:

16. Write a Python program to check whether the given number is positive or negative using a simple if statement.
17. Write a Python program to find the largest of the given three numbers using a selection statement.
18. Write a Python program to display the corresponding day of a given number in a week.
19. Write a Python program to check whether the given year is a leap year or not.

Week – 6:

20. Write a Python program to print natural numbers up to a given range.
21. Write a Python program to print the factors of a given number.
22. Write a Python program to check whether the given number is Armstrong or not.

Week – 7:

23. Write a Python program to print numbers that are divisible by a given number up to a given range using unconditional statements.
24. Write a Python program to create a list and print all elements of the list using the list method.

Week – 8:

25. Write a Python program demonstrating the following operations on a tuple:
    i) Create a tuple and copy the contents of the list.
    ii) Delete elements from the tuple after displaying the contents.
26. Write a Python program demonstrating the following operations on a dictionary:
    i) Create a dictionary and display the contents.
    ii) Create a dictionary having multiple values for the same key.
    iii) Concatenate two lists and convert them into a dictionary

Week – 9:

27. Write a Python program to create a single-dimensional array and display its contents.
28. Write a Python program implementing the following operations on arrays using the array module:
    i) Print the first n elements
    ii) Print the elements with the given stride value
    iii) Print the elements in reverse order.
    iv) Copy the contents into another array.

Week – 10:

29. Write a Python program implementing the following operations on arrays using the NumPy module:
    i) Create a 2x3 array and add elements
    ii) Convert the dimensions of the array into 3x2.
    iii) Convert the dimensions of the array into 1x6.
    iv) Copy the contents into another array.
30. Write a Python program to perform Matrix addition and Matrix multiplication.

Week – 11:

31. Write a Python program to find the sum of n natural numbers using user-defined functions.
32. Write a Python program having a function that takes a list and returns a sorted list.

Week – 12:

33. Write a Python program to print the factorial of a given number using recursive function.
34. Write a Python program to display the Fibonacci series up to a given range using functions:
    i) Without using recursion ii) Using recursion

Week – 13:

35. Write a Python program illustrating the occurrence of an exception.
36. Write a Python program to create a new file and add the given text to the file.

Week – 14:

37. Write a Python program demonstrating command-line arguments.
38. Write a Python program to add the contents into an existing file and display the contents of a file from the specified position.

Textbooks:

1. R. Nageswara Rao, “Core Python Programming”, Dreamtech
2. Fundamentals of Python First Programs, Kenneth A. Lambert, 2nd Edition, Cengage.

Reference Books:

1. Allen B. Downey, “Think Python: How to Think like a Computer Scientist”, 2nd edition, Updated for Python 3, Shroff/O‘Reilly Publishers, 2016.
2. Core Python Programming, W.Chun, Pearson.

Course Outcomes:
After completion of the Course, the student should be able to

CO-1. Apply Python syntax, operators, and I/O features to write and execute programs.
CO-2. Apply operators, control statements, and loops to solve computational problems.
CO-3. Use data structures like lists, tuples, dictionaries, and arrays to develop Python programs.
CO-4. Develop modular solutions using functions and recursion.
CO-5. Implement file operations, exception handling, and command-line arguments in Python.

Course Articulation Matrix:

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**1**	**2**
CO-1	3	2	1	-	2	-	-	-	-	1	2	-	3	1
CO-2	3	3	2	-	3	-	-	-	-	1	2	-	3	2
CO-3	3	3	3	2	3	1	-	-	-	2	3	-	3	3
CO-4	2	3	3	2	3	1	-	-	-	2	3	-	3	3
CO-5	2	3	3	2	3	2	-	-	-	2	3	-	3	3

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

I B.Tech. II Semester

Course Name	:	**ELECTRICAL CIRCUITS – II LABORATORY**				
Course Code	:	**2502204**	L	T	P	C
Course Category	:	**Laboratory**	0	0	2	1

CIE: 40 SEE: 60 Total: 100

Course Objectives:

- To design and analyze the electrical systems.
- To draw and understand the locus diagrams.
- To determine the two port network parameters.
- To analyze the time response of first order RL, RC and RLC circuits.
- To analyze the performance of filters.

The following experiments are required to be conducted as compulsory

1. To draw the current locus Diagrams of RL (R-Varying) and RC (R-Varying) Series Circuits.
2. Determination of Z- parameters.
3. Determination of ABCD parameters.
4. Frequency domain analysis of Low-pass filter.
5. Frequency domain analysis of Band-pass filter.
6. Transient Response of Series RL and RC circuits for sinusoidal excitation.
7. Determination of Hybrid parameters.
8. Determination of Time response of first order RL and RC circuit for periodic non – sinusoidal inputs - Time Constant and Steady state error.

In addition to the above eight experiments, at least any two of the experiments from the following list are required to be conducted

1. Frequency domain analysis of High-pass filter.
2. To draw the locus Diagrams of RL (L-Varying) and RC (C-Varying) Series Circuits.
3. Determination of Time response of first order RLC circuit for periodic non-sinusoidal inputs - Time Constant and Steady state error.
4. Determination of Y- parameters.

Course Outcomes: After completion of the course, the student should be able to

CO-1. Measure the three phase Active and Reactive power of electrical network
CO-2. Draw the locus diagrams of electrical circuits with R, L &C parameters.
CO-3. Analyze the performance of two port network parameters.
CO-4. Analyze the time response of first order RL, RC and RLC circuits.
CO-5. Perform frequency domain analysis of low-pass, high-pass, and band-pass filters.

Course Articulation Matrix:

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**1**	**2**
CO-1	2	1	2	2	2	-	-	1	-	1	2	3	2	-
CO-2	2	1	2	2	2	-	-	1	-	1	2	2	-	2
CO-3	2	1	1	2	2	-	-	1	-	1	2	3	1	-
CO-4	2	1	2	1	2	-	-	1	-	1	L	3	-	1
CO-5	2	1	2	2	2	-	-	1	-	1	2	3	-	2

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

I B.Tech. II Semester

Course Name	:	**ENGLISH LANGUAGE AND COMMUNICATION SKILLS LABORATORY**				
Course Code	:	**2599207**	L	T	P	C
Course Category	:	**Laboratory**	0	0	2	1

CIE: 40 SEE: 60 Total: 100

Course Objectives:

- To enable students, develop their active listening skills
- To equip students with necessary training in listening, so that they can comprehend the speech of people from different linguistic backgrounds
- To improve their pronunciation and neutralize accent
- To enable students express themselves fluently and appropriately
- To practice speaking in social and professional contexts

.

Exercise – I **12 hrs**

CALL Lab:

Instruction: Speech Sounds-Listening Skill - Importance – Purpose - Types- Barriers- Active Listening

Practice: Listening to Distinguish Speech Sounds (Minimal Pairs) **-** *Testing Exercises*

ICS Lab:

❖ **Diagnostic Test – Activity titled '*Express Your View*'**

Instruction: Spoken and Written language - Formal and Informal English - Greetings - Introducing Oneself and Others

Practice: Any Ice-Breaking Activity

Exercise – II **12 hrs**

CALL Lab:

Instruction: Listening vs. Hearing – Listening for Sequence

Practice: Listening for General Information - Multiple Choice Questions - *Listening Comprehension Exercises*

ICS Lab:

Instruction: Features of Good Conversation – Strategies for Effective Communication- Understanding Non-Verbal Communication

Practice: Role Play Activity -Situational Dialogues –Expressions used in Various Situations –Making Requests and Seeking Permissions – Taking Leave - Telephone Etiquette- Silent Speech

Exercise – III **12 hrs**

CALL Lab:

Instruction: Past Tense and Plural Markers – Tips for Neutralizing Mother Tongue Influence (MTI)

Practice: Differences between British and American Pronunciation –*Listening Comprehension Exercises*

ICS Lab:

Instruction: Describing Objects, Situations, Places, People and Events

Practice: Picture Description Activity – Looking at a Picture and Describing Objects, Situations, Places, People and Events

Exercise – IV **12 hrs**

CALL Lab:

Instruction: Techniques for *Effective* Listening

Practice: Listening for Specific Details - Gap Fill Exercises - Past Tense and Plural Markers words- *Listening Exercise*

ICS Lab:

Instruction: How to Tell a Good Story - Story Star- Sequencing-Creativity

Practice: Activity on Telling and Retelling Stories - Collage

Exercise – V **13 hrs**

CALL Lab:

Instruction: Identifying the literal and implied meaning

Practice: Listening for Evaluation - Write the Summary – Listening Comprehension Exercises

ICS Lab:

Instruction: Group Discussion Skills

Practice: Mock Group Discussions-Debates

❖ **Post-Assessment Test on 'Express Your View'**

Minimum Requirement of infrastructural facilities for ELCS Lab:

1. Computer Assisted Language Learning (CALL) Lab:

The Computer Assisted Language Learning Lab has to accommodate 40 students with 40 systems, with one Master Console, LAN facility and English language learning software for self- study by students.

System Requirement (Hardware component):

Computer network with LAN facility (minimum 40 systems with multimedia) with the following specifications:

i) Computers with Suitable Configuration
ii) High Fidelity Headphones

2. Interactive Communication Skills (ICS) Lab:

The Interactive Communication Skills Lab: A Spacious room with movable chairs and audio-visual aids with a Public Address System, LCD etc.

Suggested Software:

- Cambridge Advanced Learners' English Dictionary with CD.
- Grammar Made Easy by Darling Kindersley.
- Punctuation Made Easy by Darling Kindersley.
- Oxford Advanced Learner's Compass, 10th Edition.
- English in Mind (Series 1-4), Herbert Puchta and Jeff Stranks with Meredith Levy, Cambridge.
- English Pronunciation in Use (Elementary, Intermediate, Advanced) Cambridge University Press.
- English Vocabulary in Use (Elementary, Intermediate, Advanced) Cambridge University Press.
- TOEFL & GRE (KAPLAN, AARCO & BARRONS, USA, Cracking GRE by CLIFFS).

References:

1. Shobha, KN &Rayen, J. Lourdes. (2019). *Communicative English – A workbook.* Cambridge University Press
2. Board of Editors. (2016). *ELCS Lab Manual: A Workbook for CALL and ICS Lab Activities.* Orient BlackSwanPvt. Ltd.
3. Mishra, Veerendra et al. (2020). *English Language Skills: A Practical Approach.* Cambridge University Press
4. (2022). *English Language Communication Skills – Lab Manual cum Workbook.* Cengage Learning India Pvt. Ltd.
5. Ur, Penny and Wright, Andrew. 2022. *Five Minute Activities – A Resource Book for Language Teachers.* Cambridge University Press.

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Course Outcomes: After completion of the course, the student should be able to

CO-1. Listen actively and identify important information in spoken texts.
CO-2. Interpret the speech and infer the intention of the speaker
CO-3. Improve their accent for intelligibility
CO-4. Speak fluently with clarity and confidence
CO-5. Use the language in real life situations

Course Articulation Matrix:

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO-1	-	-	-	-	-	1	3	-	2	3	-	-	-	2
CO-2	-	-	-	-	-	3	3	-	3	2	-	-	-	2
CO-3	-	-	-	-	-	2	2	1	2	3	-	-	-	1
CO-4	-	-	-	-	-	2	2	2	2	2	-	-	-	2
CO-5	-	-	-	-	-	2	2		3	3	-	-	-	2

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

I B.Tech. II Semester

Course Name	:	**ENGINEERING WORKSHOP**				
Course Code	:	**2503202**	L	T	P	C
Course Category	:	**Laboratory**	0	0	2	1

CIE: 40 SEE: 60 Total: 100

Course Objectives:

- To introduce students to basic manufacturing processes and workshop practices.
- To provide hands-on training in carpentry, fitting, welding, sheet metal, and machining
- To develop skills in using and tools and measuring instruments.
- To enhance safety awareness and proper handling of workshop equipment.
- To build a foundational understanding of industrial production and fabrication.

1. TRADES FOR EXERCISES:

At least two exercises from each trade:

i. **Carpentry:** (T-Lap Joint, Dovetail Joint, Mortise & Tenon Joint).
ii. **Fitting:** (V-Fit, Dovetail Fit & Semi-circular fit).
iii. **Tin-Smithy:** (Square Tin, Rectangular Tray & Conical Funnel).
iv. **Foundry:** (Preparation of Green Sand Mould using Single Piece and Split Pattern).
v. **Welding Practice:** (Arc Welding & Gas Welding).
vi. **House-wiring:** (Parallel & Series, Two-way Switch and Tube Light).
vii. **Black Smithy:** (Round to Square, Fan Hook and S-Hook).

2. TRADES FOR DEMONSTRATION & EXPOSURE:

I. Plumbing, Machine Shop, Metal Cutting, Power tools in construction and Wood Working.

Textbooks:

1. Workshop Practice, B. L. Juneja, Cengage Learning India,1st edition,2015.
2. Workshop Manual, K.Venkata Reddy, BS Publication, 6thEdition, Rpt.2025.

References:

1. Workshop Manual, K.Venugopal, Anuradha Publications,2012thedition,2012.

Course Outcomes: After completion of the lab, the student should be able to

CO-1. Understand the basic manufacturing processes and operations.
CO-2. Use hand tools and equipment safely and efficiently.
CO-3. Perform basic operations in carpentry, fitting, welding, sheet metal work and machining
CO-4. Read and interpret work shop drawings
CO-5. Develop teamwork, time management and quality awareness in a workshop environment.

Course Articulation Matrix:

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	1	2
CO-1	1	-	1	-	-	-	-	-	2	-	-	-	1	-
CO-2	1	-	-	-	-	-	-	-	2	-	-	-	-	-
CO-3	2	-	-	-	-	-	-	-	-	-	1	-	2	-
CO-4	-	-	-	-	-	-	-	-	-	-	1	-	-	-
CO-5	1	-	-	-	-	-	-	-	-	-	1	-	2	-

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

II Year I Semester

II B.Tech. I Semester

Course Name	:	**ELECTROMAGNETIC FIELDS**				
Course Code	:	**2502301**	L	T	P	C
Course Category	:	**Theory**	3	0	0	3

CIE: 40 SEE: 60 Total: 100

Course Objectives:

- To introduce the concepts of Electric field.
- To introduce the concepts of Dielectrics and Capacitance.
- Ability to analyze the Ampere's circuital law and its applications.
- Applications of electric and magnetic fields in the development of the theory for power transmission lines and electrical machines.
- To Study about Electromagnetic waves.

UNIT-I: **12 Hrs**

Static Electric Field: Review of conversion of a vector from one coordinate system to another coordinate system, Coulomb's law, Electric field intensity, Electrical field due to point charges. Line, Surface and Volume charge distributions. Gauss law and its applications, Electric dipole, Electrostatic Energy and Energy density

UNIT-II: **12 Hrs**

Conductors, Dielectrics And Capacitance: Current and current density, Ohms Law in Point form, Continuity equation, Boundary conditions of conductors and dielectric materials. Capacitance, Capacitance of a two-wire line, Poisson's equation, Laplace's equation, Solution of Laplace's equation in one variable.

UNIT-III: **13 Hrs**

Static Magnetic Fields and Magnetic Forces: Biot-Savart Law, Ampere Circuital Law, Magnetic flux and Magnetic flux density, Magnetic field intensity (MFI), Scalar and Vector Magnetic potentials, Steady magnetic fields produced by current carrying conductors

UNIT-IV: **13 Hrs**

Force on a Moving Charge: Lorentz force equation, Force on a differential current element, Force between two straight long and parallel current carrying conductors, Magnetic Circuits, Self-Inductances and Mutual Inductances.

UNIT-V: **12 Hrs**

Time Varying Fields And Maxwell's Equations: Faraday's laws of Electromagnetic induction, Displacement current, Point form of Maxwell's equation, Integral form of Maxwell's equations, Poynting theorem.

Textbooks:

1. Electromagnetic Fields - Sadiku, Oxford Publications, 8th Edition, 2014.
2. Engineering Electromagnetics - William H. Hayt & John. A. Buck, McGraw Hill Companies, 8th Editon, 2014.

References:

1. Introduction to Electro Dynamics - D J Griffiths, Prentice-Hall of India Pvt. Ltd, 4th Edition, 2012.
2. Electromagnetic - J P Tewari, Khanna Publishers, 2nd Edition, 2005.
3. Electromagnetics - J. D Kraus, McGraw Hill Inc, 6th Edition, 2012.
4. Electromagnetic fields - S. Kamakshaiah, Right Publishers, 1st Edition, 2007.

Course Outcomes: After completion of the course, the student should be able to

CO-1. Understand the basic laws of electromagnetism and their applications. Electromagnetic problems.
CO-2. Understand the Properties and behavior of conductors, dielectrics & Capacitance.
C0-3. Magneto statics and Physical laws of electro magnetism and Force in magnetic fields.
CO-4. Analyze time varying fields in point form and Integral form.
C0-5. Understand the Propagation of EM Waves

Course Articulation Matrix:

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**1**	**2**
CO-1	1	2	3	3	3	2	3	-	-	-	2	2	2	2
CO-2	2	2	3	3	3	2	3	-	-	-	2	2	3	3
CO-3	1	2	3	3	3	2	3	-	-	-	2	2	3	3
CO-4	2	2	3	2	3	2	-	-	-	-	2	2	3	3
CO-5	3	2	2	2	3	2	2	-	-	-	2	2	3	3

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

II B.Tech. I Semester

Course Name	:	**ELECTRICAL MACHINES-I**				
Course Code	:	**2502302**	L	T	P	C
Course Category	:	**Theory**	3	0	0	3

CIE: 40 SEE: 60 Total: 100

Course Objectives:

- To study and understand different types of DC machines and their performance evaluation through various testing methods.
- To study and understand principle of DC motors & Testing of DC machines
- To study and understand principle of 1-phase transformer
- To study and understand performance & Testing of 1-phase transformer
- To study and understand performance & Testing of 3-phase transformer

UNIT-I: **13 Hrs**

D.C. Generators: Principle of operation – Action of commutator – constructional features – armature windings – lap and wave windings – simplex and multiplex windings (elementary treatment only) – EMF Equation. Concept of Armature reaction and commutation – Cross magnetizing and de-magnetizing AT/pole. Methods of Excitation – separately excited and self-excited generators – build-up of EMF - critical field resistance and critical speed. Performance Characteristics of shunt, series and compound generators and applications.

UNIT-II: **12 Hrs**

DC Motors: Principle of operation – Back EMF. - Torque equation – characteristics and application of shunt, series and compound motors.

3-point starter, Speed control of DC shunt and series motors - Armature voltage and field flux control methods. Losses – Constant & Variable losses –calculation of efficiency – condition for maximum efficiency.

Testing of DC Machines: Methods of Testing – Direct, Indirect, and Regenerative Testing – Brake Test – Swinburne's Test – Hopkinson's Test.

UNIT-III: **12 Hrs**

Single Phase Transformers: Types - constructional details-minimization of hysteresis and eddy current losses- EMF equation - operation on no-load and on load - phasor diagrams and Applications.

UNIT-IV: **13 Hrs**

Equivalent circuit - losses and efficiency – regulation - All day efficiency - effect of variations of frequency & supply voltage on iron losses**.**

Testing of Transformers: Open Circuit and Short Circuit tests - Sumpner's Test - predetermination of efficiency and regulation-separation of losses test..

UNIT-V: **13 Hrs**

Parallel operation with equal and unequal voltage ratios - auto transformers-equivalent circuit - comparison with two winding transformers.

Poly-phase transformers – Poly-phase connections - Y/Y, Y/Δ, Δ/Y, Δ/Δ and open Δ, Scott connection and Applications.

Textbooks:

1. Theory and performance of Electrical machines – J.B Gupta, S.K Kataria & Sons publishers, 2009.
2. Electric Machinery- P.S. Bimbra, Khanna Publishers, 7th edition, 2010.

References:

1. Electrical Machines - I.J. Nagrath & Kothari, McGraw Hill Companies, 3rd edition, 2004.
2. A Text book of Electrical Technology – B.L. Theraja and A.K. Theraja, S. Chand Publications, Vol2 eprint 2015.
3. Electrical Machines – S.K. Bhattacharya, McGraw Hill Companies, 2007.
4. Electric Machines – M.V. Deeshpande, Wheeler Publishing, 1997..

Course Outcomes:

CO-1.After completion of the course, the student should be able to identify different parts of a DC machines & understand their operation.
CO-2.This course provides characteristics of DC motor, Testing & applications of DC machines.
CO-3.This course provides the principle of 1-phase transformer
CO-4.This course analyzes the testing & performance of 1-phase transformer.
CO-5.This course analyzes the testing & performance of 3-phase transformer.

Course Articulation Matrix:

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**1**	**2**
CO-1	1	-	3	-	2	-	-	-	-	-	1	1	1	2
CO-2	1	-	-	-	2	-	-	-	-	-	1	1	2	1
CO-3	1	-	3	-	2	-	-	-	-	-	1	1	1	1
CO-4	1	-	3	-	2	-	-	-	-	-	1	1	2	1
CO-5	1	-	3	-	2	-	-	-	-	-	1	1	2	2

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

II B.Tech. I Semester

Course Name	:	**ELECTRONIC DEVICES AND CIRCUITS**				
Course Code	:	**2504301**	L	T	P	C
Course Category	:	**Theory**	3	0	0	3

CIE: 40 SEE: 60 Total: 100

Course Objectives:

- To introduce components such as diodes and know the applications of Diode.
- To understand the characteristics of the BJT.
- To Learn the concepts of, load line analysis, biasing techniques
- To analyze the transistor amplifier using h-parameter model
- To understand the characteristics of the JFET and MOSFET.

UNIT-I: **12Hrs**

Diode Characteristics and Applications: PN junction diode – I-V characteristics, Diode resistance and capacitance, Rectifiers — Half-wave, Full- wave (Center-tap and bridge), Clippers and clampers, Zener diode – I-V characteristics and voltage regulation.

UNIT-II: **12Hrs**

Bipolar Junction Transistor (BJT): Structure and working principle of BJT, Current components and transistor action, Configurations: Common Base (CB), Common Emitter (CE), Common Collector (CC), Input and output characteristics, Determination of h-parameters from transistor characteristics.

UNIT-III: **12Hrs**

BJT Biasing: Need for biasing and stabilization, Load line and operating point, Biasing techniques: Fixed bias, Collector-to-base bias, Voltage divider bias, Stability factors and thermal runaway

UNIT-IV: **12Hrs**

Transistor Amplifiers: Transistor as a small-signal amplifier, h-parameter equivalent circuit, CE, CB, CC amplifier analysis using h-parameters, Approximate CE model – with and without emitter bypass capacitor.

UNIT-V: **12Hrs**

Special Purpose Diodes: Principle of Operation of — SCR, Tunnel Diode, Varactor Diode,
Field Effect Transistors and Advanced Devices: JFET: Structure, operation, and characteristics, MOSFET: Enhancement and Depletion modes — Structure, operation, and characteristics.

Textbooks:

1. Millman, Jacob, and Christos C. Halkias. *Electronic Devices and Circuits*. Tata McGraw-Hill, 1991.
2. Boylestad, Robert L., and Louis Nashelsky. *Electronic Devices and Circuit Theory*. Pearson, 11th ed., 2013.
3. Sedra, Adel S., and Kenneth C. Smith. *Microelectronic Circuits*. Oxford University Press, 7th ed., 2014.

References:

1. Bell, David A. *Electronic Devices and Circuits*. Oxford University Press, 5th ed., 2008.
2. Neamen, Donald A. *Electronic Circuit Analysis and Design*. McGraw-Hill, 2nd ed., 2001.
3. Salivahanan, S., and N. Suresh Kumar. *Electronic Devices and Circuits*. McGraw-Hill Education, 4th ed., 2017.
4. Razavi, Behzad. *Fundamentals of Microelectronics*. Wiley, 2nd ed., 2013.
5. Taur, Yuan, and Tak H. Ning. *Fundamentals of Modern VLSI Devices*. Cambridge University Press, 2nd ed., 2009.

Course Outcomes:

CO1: Analyze the PN Junction diode operation and its characteristics
CO2: Analyze the characteristics of BJT.
CO3: Design the amplifiers with various biasing techniques
CO4: Analyze single stage amplifiers using hybrid model.
CO5: Analyze the characteristics of FET

Course Articulation Matrix:

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**1**	**2**
CO-1	3	1	2	-	-	1	1	-	-	-	-	1	2	-
CO-2	3	2	3	-	-	2	1	-	-	-	-	1	3	-
CO-3	2	3	3	2	-	-	-	-	-	-	-	1	-	1
CO-4	2	3	3	2	-	-	-	-	-	-	-	1	-	1
CO-5	3	2	3	-	-	2	1	-	-	-	-	1	3	-

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

II B.Tech. I Semester

Course Name	:	**POWER SYSTEMS–I**				
Course Code	:	**2502303**	L	T	P	C
Course Category	:	**Theory**	3	0	0	3

CIE: 40 SEE: 60 Total: 100

Course Objectives:

- To understand the power generation through conventional and non-conventional sources.
- To illustrate the economic aspects of power generation and tariff methods.
- To know about fundamental concepts of air insulated substations
- To give an idea about fundamental concepts of gas insulated substations.
- To understand the concepts of AC distribution systems.

UNIT-I: 13 Hrs

Generation of Electric Power: Operation of Hydel, Thermal, Nuclear and Gas Power plant with layouts - Description of components-Choice of site - advantages and disadvantages, Introduction and description of components- renewable energy sources and plants (solar and wind).

UNIT-II: 12 Hrs

Economics of Power Generation: Introduction, definitions of connected load, maximum demand, demand factor, load factor, diversity factor, Load curve, Load duration curve, number and size of generator units.

Base load and peak load plants.

Cost of electrical energy-fixed cost, running cost, Tariffs.

UNIT-III: 12 Hrs

Air Insulated Substations (AIS): Indoor & Outdoor substations: Substations layout showing the location of all the substation equipment. Bus bar arrangements in the Sub-Stations: Simple arrangements like single bus bar, sectionalized single bus bar, main and transfer bus bar system with relevant diagrams.

UNIT-IV: 11 Hrs

Gas Insulated Substations (GIS): Advantages of Gas insulated substations, different types of gas insulated substations, single line diagram of gas insulated substations, bus bar, construction aspects of GIS, Installation and maintenance of GIS, Comparison of Air insulated substations and Gas insulated substations.

UNIT-V: 12 Hrs

AC Distribution: Introduction, AC distribution, Single phase, 3-phase 3 wire, 3-phase 4 wire system, bus bar arrangement, Selection of site for substation. Voltage Drop Calculations (Numerical Problems) in AC Distributors for the following cases: Power Factors referred to receiving end voltage and with respect to respective load voltages.

Textbooks:

1. C. L. Wadhwa, “Generation, transmission and distribution of electrical power”, 3rd Edition, 2017.
2. A. Chakrabarti, M.L. Soni, P.V. Gupta, U.S. Bhatnagar, “A Text book on Power System Engineering”, Dhanpat Rai Publishing Company (P) Ltd, Paperback Edition, 2016.
3. J. B. Gupta, A Course in Power Systems” Katson Books, 11th Edition, 2016.

References:

1. C.L.Wadhwa, “Electrical Power Systems”, 8th Edition, New Age International, 2022.
2. V. K. Mehta and Rohit Mehta, “Principles of Power Systems”, S. Chand & Company Ltd, New Delhi, 2004.

Course Outcomes: After completion of the course, the student should be able to

CO-1.Understand the power generation through conventional and non-conventional sources
CO-2.Evaluate the power tariff methods and Economics associated with power generation
CO-3.Understand layout of air insulated substations, their equipment’s functioning
CO-4.Analyze the operations of GIS
CO-5.Calculate the voltage drop in different A.C Distributors

Course Articulation Matrix:

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO-1	3	-	2	-	-	1	-	2	2	2	-	-	2	2
CO-2	2	-	3	-	2	-	-	3	2	1	-	2	3	2
CO-3	3	2	2	-	2	-	-	1	-	-	2	-	2	2
CO-4	2	2	2	-	3	-	-	-	2	-	2	-	3	3
CO-5	2	2	2	-	3	-	-	-	1	-	3	-	3	3

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

II B.Tech. I Semester

Course Name	:	**ELECTRICAL MEASUREMENTS AND SENSORS**				
Course Code	:	**2502304**	L	T	P	C
Course Category	:	**Theory**	2	0	0	2

CIE: 40 SEE: 60 Total: 100

Course Objectives:

- To introduce the basic principles of all measuring instruments.
- To understand the principle of operation of potentiometers and instrument Transformers.
- To deal with the measurement of power, energy.
- To deal with the R, L, C measuring bridges.
- To understand the basic concepts of smart and digital metering.

UNIT-I: 12 Hrs

Introduction to Measuring Instruments: Classification – deflecting, control and damping torques – Ammeters and Voltmeters – PMMC, moving iron type instruments – expression for the deflecting torque and control torque – Errors and compensations, extension of range using shunts and series resistance. Electrostatic Voltmeters-electrometer type and attracted disc type – extension of range of Electrostatic Voltmeters.

UNIT-II: 12 Hrs

Potentiometers & Instrument Transformers: Principle and operation of DC Crompton's potentiometer – standardization – Measurement of unknown resistance, current, voltage. AC Potentiometers: polar and coordinate type's standardization – applications. CT and PT – Ratio and phase angle errors (Qualitative approach).

UNIT-III: 12 Hrs

Measurement of Power & Energy: Single phase dynamometer wattmeter, LPF and UPF, Double element and three element dynamometer wattmeters, expression for deflecting and control torques – Extension of range of wattmeter using instrument transformers – Measurement of active and reactive powers in balanced and unbalanced systems.

Single phase induction type energy meter – driving and braking torques – errors and compensations.

UNIT-IV: 12 Hrs

DC & AC Bridges: Method of measuring low, medium and high resistance – sensitivity of Wheat-stone's bridge – Kelvin's double bridge for measuring low resistance, measurement of high resistance – loss of charge method.

Measurement of inductance- Maxwell's bridge, Hay's bridge, Anderson's bridge. Measurement of capacitance and loss angle –De Sauty's Bridge - Wien's bridge – Schering Bridge. (Qualitative approach)

UNIT-V: 12 Hrs

Sensors- Classification of transducers- Temperature sensors- Proximity sensor- Pressure sensor- IR sensors- Motion detection sensors- Ultrasonic sensors- Rotor Position Sensors, Operation of Strain Gauge- Thermocouples, construction and working of LVDT, Piezo electric transducers, photovoltaic, photo conductive cells, and photo diodes-Applications.

Smart instruments: Intelligent transducer, self-diagnosis and remote calibration features, HART communication, MEMS, non-linearity compensation; smart energy meter components, working principle; Automatic Meter Reading (AMR), Advanced Metering Infrastructure (AMI) environments

Textbooks:

1. A.K. Sawhney, “A course in Electrical & Electronic Measurement Instrumentation”, Dhanpat Rai & Co. Publications, 2012.
2. Dr.Rajendra Prasad, “Electrical Measurements & Measuring Instruments”, Khanna Publishers 2019.

References:

1. G. K. Banerjee, “Electrical and Electronic Measurements”, PHI Learning Pvt. Ltd., 2nd Edition, 2016.
2. R.K.Rajput, “Electrical & Electronic Measurement & Instrumentation”, S.Chand and Company Ltd., 4th Edition, 2016.
3. S.C.Bhargava, “Electrical Measuring Instruments and Measurements”, BS Publications, 2012

Course Outcomes: After completion of the course, the student should be able to

CO-1: Understand the working of different types of ammeters and voltmeter, their construction, operation, errors and compensation
CO-2: Analyze the instrument transformers, errors and understanding the construction and operation of DC & AC potentiometer
CO-3: Develop knowledge on LPF&UPF wattmeter and their extension, working of energy meters
CO-4: Acquire the knowledge on measuring R, L, C parameters.
CO-5: Understanding and analyzing the sensors and smart meters.

Course Articulation Matrix:

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**1**	**2**
CO-1	3	-	2	-	2	-	-	-	1	-	2	-	3	2
CO-2	3	2	2	-	3	-	-	-	1	-	3	-	3	3
CO-3	2	2	2	-	2	-	-	-	2	-	2	-	3	2
CO-4	3	2	1	-	2	-	-	-	1	-	2	-	3	2
CO-5	2	2	2	-	3	-	-	1	2	2	3	-	2	3

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

II B.Tech. I Semester

Course Name	:	**INNOVATION AND ENTREPRENEURSHIP**				
Course Code	:	**2588301**	L	T	P	C
Course Category	:	**Theory**	2	0	0	2

CIE: 40 SEE: 60 Total: 100

Course Objectives:

- To familiarize on the basic concepts of innovation, entrepreneurship and its importance.
- To Identify and analyze the process of problem-opportunity identification, market segmentation, and idea generation techniques.
- To initiate prototype development and understand minimum viable product.
- To develop initial Business and financial planning and Go-to-Market strategies.
- To impart knowledge on establishing startups, venture pitching and IPR.

Unit I: Fundamentals of Innovation and Entrepreneurship **12 Hrs**

Innovation: Introduction, need for innovation, Features, Types of innovations, innovations in manufacturing and service sectors, fostering a culture of innovation, planning for innovation.

Entrepreneurship: Introduction, types of entrepreneurship attributes, mindset of entrepreneurial and intrapreneurial leadership, Role of entrepreneurs in economic development. Woman Entrepreneurship, Importance of on-campus startups. Understanding to build entrepreneurial mindset attributes and networks individuals while on campus.

Core Teaching Tool: Simulation, Game, Industry Case Studies (Personalized for students – 16 industries to choose from), Venture Activity.

Unit II: Problem and Customer Identification **12 Hrs**

Identification of gap, problem, analyzing the problem from a industry perspective, real-world problems**,** market and customer segmentation, validation of customer problem fit, Iterating problem-customer fit, Competition and Industry trends mapping and assessing initial opportunity, Porter's Five Force Model. Idea generation, Ideation techniques: Brainstorming, Brain writing, Round robin, and SCAMPER, Design thinking principles, Mapping of solution to problem.

Core Teaching Tool: Several types of activities including: Class, game, Gen AI, 'Get out of the Building' and Venture Activity.

Unit III: Opportunity assessment and Prototype development **12 Hrs**

Identify and map global competitors, review industry trends, and understand market sizing: TAM, SAM, and SOM. Assessing scope and potential scale for the opportunity.

Understanding prototyping and Minimum Viable Product (MVP). Developing a prototype: Testing, and validation.

Core Teaching Tool: Venture Activity, no-code Innovation tools, Class activity

Unit IV: Business & Financial Models **12 Hrs**

Introduction to Business Model and types, Lean Canvas Approach: 9-block lean canvas model, building lean canvas for your startup. Business planning: components of Business plan- Sales plan, People plan and financial plan, Financial Planning: Types of costs, preparing a financial plan for profitability using a financial template, understanding the basics of Unit economics, Economies of Scale and analyzing financial performance. Go-To-Market (GTM) approach – Selecting the Right Channel, creating digital presence, and building customer acquisition strategy.

Core Teaching Tool: Founder Case Studies – Sama and Securely Share; Class activity and discussions; Venture Activities.

Unit V: Startups and IPR **12 Hrs**

Startup requirements, building founding team members and mentors, pitch preparation, start-up registration process, funding opportunities and schemes, institutional support to entrepreneurs, startup lifecycle, documentation, legal aspects in startup, venture pitching readiness, National Innovation Startup Policy (NISP) and its features.

Patents, Designs, Patentability, Procedure for grants of patents. Indian Scenario of Patenting, International Scenario: International cooperation on Intellectual Property. Patent Rights: Scope of Patent Rights. Copyright, trademark, and GI. Licensing and transfer of technology.

Core Teaching Tool: Expert talks; Cases; Class activity and discussions; Venture Activities.

Suggested Readings:

1. John R Bessant, Joe Tidd, Innovation and Entrepreneurship, 4E, Wiley, Latest Edition.
2. Ajay Batra, The Stratup Launch Book- A Practical Guide for Launching Customer Centric Ventures, Wiley, 2020. (For Core Teaching Tool).
3. Entrepreneurship Development and Small Business Enterprises, Poornima M Charantimath, 3E, Pearson, 2018.
4. D.F. Kuratko and T.V. Rao, Entrepreneurship: A South-Asian Perspective, Cengage Learning, 2013.
5. Robert D. Hisrich, Michael P. Peters, Dean A. Shepherd, Sabyasachi Sinha (2020). Entrepreneurship, McGrawHill, 11th Edition.
6. NISP -Brochure inside pages - startup_policy_2019.pdf

Course Outcomes: After completion of the course, the student should be able to

CO-1: Understand the entrepreneurship and the entrepreneurial process and its significance in economic development

CO-2: Assess the problem from an industry perspective and generate solutions using the design thinking principles.

CO-3: Assess market competition, estimate market size, and develop a prototype

CO-4: Analyze Business and financial planning models and Go-to-Market strategies.

CO-5: Able to build a start-up, register IP and identify funding opportunities.

Course Articulation Matrix:

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO-1	2	-	-	-	2	3	2	3	2	3	-	-	2	3
CO-2	3	3	3	2	3	2	-	2	2	2	3	-	2	3
CO-3	2	2	3	2	3	2	-	3	2	2	3	-	2	3
CO-4	2	-	2	2	3	2	-	3	2	3	2	3	1	3
CO-5	2	-	2	2	3	3	-	3	2	3	2	3	1	3

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

II B.Tech. I Semester

Course Name	:	**ELECTRICAL MACHINES-I LABORATORY**				
Course Code	:	**2502305**	L	T	P	C
Course Category	:	**Laboratory**	0	0	2	1

CIE: 40 SEE: 60 Total: 100

Course Objectives:

- To give students an insight into the construction details of DC machine.
- To train students to perform tests on DC generator.
- To train the students to analyze DC Generator.
- To train students to perform tests on DC Motor.
- To examine the performance of Single and Three Phase Transformers.

The following experiments are required to be conducted compulsory experiments:

1. Magnetization characteristics of DC shunt generator (Determination of critical field resistance and critical speed)
2. Load test on DC shunt generator (Determination of characteristics)
3. Load test on DC series generator (Determination of characteristics)
4. Hopkinson's test on DC shunt machines (Predetermination of efficiency)
5. Swinburne's test and speed control of DC shunt motor (Pre-determination of efficiencies)
6. Brake test on DC compound motor (Determination of performance curves)
7. OC and SC Test on Single Phase Transformer
8. Three Phase Transformer: Verification of Relationship between Voltages and Currents (Star- Delta, Delta-Delta, Delta-star, Star-Star)

In addition to the above eight experiments, atleast any two of the experiments from the following list are required to be conducted:

1. Brake test on DC shunt motor (Determination of performance curves)
2. Load test on DC compound generator (Determination of characteristics.
3. Fields test on DC series machines (Determination of efficiency).
4. Retardation test on DC shunt motor (Determination of losses at rated speed)
5. Separations of losses in DC shunt motor.
6. Load Test on Single Phase Transformer (Calculate Efficiency and Regulation)

Course Outcomes: After completion of the course, the student should be able to

CO-1.Interpret the constructional details of the DC machine.
CO-2.Test the performance of the DC generator
CO-3.Analyze the performance of the DC generator.
CO-4.Test and analyze the performance of the DC motor.
CO-5.Evaluate the performance of different Transformers using different testing methods.

Course Articulation Matrix:

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**1**	**2**
CO-1	1	-	3	-	2	-	-	-	-	-	1	1	1	1
CO-2	1	-	-	-	2	-	-	-	-	-	1	1	2	1
CO-3	1	-	3	-	2	-	-	-	-	-	1	1	3	1
CO-4	1	-	3	-	2	-	-	-	-	-	1	1	2	1
CO-5	1	-	3	-	2	-	-	-	-	-	1	1	2	1

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

II B.Tech. I Semester

Course Name	:	**ELECTRICAL MEASUREMENTS AND SENSORS LABORATORY**				
Course Code	:	**2502306**	L	T	P	C
Course Category	:	**Laboratory**	0	0	2	1

CIE: 40 SEE: 60 Total: 100

Course Objectives:

- To familiarize with the fundamental principles procedures for calibrating and testing various electrical measuring instruments.
- To provide hands-on experience in measuring three-phase active and reactive power using single-phase wattmeter and current transformers (CTs).
- To understand how to test and analyze instrument transformers, specifically for calculating ratio and phase angle errors.
- To introduce advanced measurement techniques for non-electrical quantities, such as using an LVDT for displacement and a strain gauge for strain.
- To enable students to apply AC and DC bridge circuits for the accurate measurement of electrical quantities like resistance, inductance, and transformer turns ratios.

The following experiments are required to be conducted as compulsory experiments:

1. Calibration and testing of single-phase energy Meter.
2. Calibration of dynamometer power factor meter.
3. Measurement of 3-phase power with single wattmeter and two CTs.
4. C.T.testing using mutual Inductor–Measurement of %ratio error and phase angle of given CT by Null method.
5. Dielectric testing of oil using HT Testing Kit.
6. Schering Bridge & Anderson Bridge.
7. Measurement of 3-Phase reactive power with single-phase wattmeter.
8. Measurement of displacement with the help of LVDT.

In addition to the above eight experiments, at least any two of the experiments from the following list are required to be conducted:

1. Calibration LPF wattmeter – by Phantom testing.
2. Crompton DC Potentiometer – Calibration of PMMC ammeter and PMMC voltmeter.
3. Kelvin's double Bridge – Measurement of resistance – Determination of Tolerance.
4. Resistance strain gauge – strain measurements and Calibration.
5. Transformer turns ratio measurement using AC bridges.

Course Outcomes: After completion of the course, the student should be able to

CO-1: Perform the calibration and testing of single-phase energy meters, dynamometer power factor meters.
CO-2: Calculate three-phase active and reactive power using a single-phase wattmeter.
CO-3: Conduct tests on current transformers (CTs) using a mutual inductor.
CO-4: Measure physical parameters like displacement and strain using transducers such as LVDTs and resistance strain gauges.
CO-5: Measure unknown resistance, inductance, and transformer turns ratio by using various AC and DC bridges.

Course Articulation Matrix:

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**1**	**2**
CO-1	3	2	-	-	2	-	1	-	1	-	3	-	3	2
CO-2	3	2	-	-	3	-	1	-	1	-	3	-	3	2
CO-3	3	3	-	-	2	-	-	-	1	-	3	-	2	3
CO-4	3	2	-	-	2	-	-	-	2	-	3	-	3	2
CO-5	3	3	-	-	3	-	-	-	2	-	3	-	3	3

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

II B.Tech. I Semester

Course Name	:	**ELECTRONIC DEVICES AND CIRCUITS LABORATORY**				
Course Code	:	**2504302**	L	T	P	C
Course Category	:	**Laboratory**	0	0	2	1

CIE: 40 SEE: 60 Total: 100

Course Objectives:

- Understand the working principles of semiconductor devices like diodes, BJTs, and FETs.
- Build and test basic rectifier, clipper, clamper, and voltage regulator circuits.
- Learn transistor biasing techniques and operating point determination.
- Analyze and design transistor amplifier circuits in various configurations.
- Use simulation software to design and validate electronic circuits.

List of experiments:

A. Hardware-Based Experiments (7):

1. Study the I–V characteristics of a PN junction diode in forward and reverse bias to determine cut-in voltage and dynamic resistance.
2. Examine the reverse bias characteristics of a Zener diode and demonstrate its application as a voltage regulator under varying conditions.
3. Design and analyze half-wave and full-wave rectifiers (center-tap and bridge) with and without capacitor filters to evaluate ripple factor and output voltage.
4. Implement clipper and clamper circuits to observe waveform shaping through positive, negative, and biased configurations.
5. Plot the input and output characteristics of a BJT in common emitter configuration to determine input/output resistance and current gain.
6. Design and test fixed bias and voltage divider bias circuits to establish a stable operating point for a BJT amplifier and study DC load line behavior.
7. Construct and analyze a Common Base (CB) configuration of a BJT to study input-output characteristics and determine current gain (α) and input/output resistance.

B. Software-Based Simulation Experiments (7):

1. Simulate a full-wave bridge rectifier with capacitor filter to analyze waveform smoothing and ripple reduction in DC power supply design.
2. Simulate a Zener diode-based voltage regulator to study voltage stabilization against varying supply voltages and load resistances.
3. Simulate a common emitter amplifier with and without emitter bypass capacitor to analyze the effect on voltage gain and signal amplification.
4. Simulate BJT operation as a switch and small-signal amplifier to understand its dual functionality in digital and analog applications.
5. Simulate the Drain and transfer characteristics of a JFET to determine parameters such as pinch-off voltage, drain resistance, and transconductance.
6. Implement clipper circuits to observe waveform shaping through positive, negative, and biased configurations.
7. Implement clamper circuits to observe waveform shaping through positive, negative, and biased configurations.

Hardware Requirements:

1. Regulated DC Power Supply (0–30V)
2. Function Generator
3. Digital Multimeter
4. Cathode Ray Oscilloscope (CRO) or DSO
5. Breadboards and Connecting Wires
6. Resistors, Capacitors, Diodes (1N4007, Zener Diodes)
7. BJTs (e.g., BC107, 2N2222), JFETs (e.g., J201), MOSFETs (e.g., IRF540N)
8. Trainer Kits (optional but preferred for ease)

Software Requirements (Any one of the listed tools or equivalent):

1. LTSpice (Free from Analog Devices)
2. NI Multisim (Academic License or Student Version)
3. Proteus Design Suite (Simulation and PCB Design)
4. TINA-TI (Free from Texas Instruments)
5. PSPICE for TI or OrCAD Lite
6. Windows PC or Laptop with minimum 4GB RAM and i3 processor or better.

Course Outcomes: After completion of the course, the student should be able to

CO1: Analyze the I–V characteristics of semiconductor devices such as diodes, BJTs, and FETs.
CO2: Design and evaluate basic rectifier, clipper, clamper, and voltage regulation circuits.
CO3: Demonstrate biasing techniques for BJTs and determine their operating point using DC load line analysis.
CO4: Design and analyze transistor amplifier circuits in various configurations using h-parameter models.
CO5: Simulate and interpret electronic circuits using appropriate simulation tools.

Course Articulation Matrix:

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**1**	**2**
CO-1	3	3	2	2	1	-	-	-	-	-	-	-	-	-
CO-2	3	3	3	2	1	1	-	-	-	-	-	-	1	-
CO-3	3	3	2	2	1	-	-	-	-	-	-	-	1	1
CO-4	3	3	3	2	1	-	-	-	-	-	-	-	2	1
CO-5	2	2	2	3	3	-	-	-	-	-	-	-	1	2

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

II B.Tech. I Semester

Course Name	:	**DESIGN OF ELECTRICAL SYSTEMS USING AUTOCAD**				
Course Code	:	**2502307**	L	T	P	C
Course Category	:	**Laboratory**	0	0	2	1

CIE: 40 SEE: 60 Total: 100

Course Objectives:

- To familiarize students with the AutoCAD interface and basic tools for electrical drafting.
- To introduce the design process of internal wiring and lighting systems in residential buildings
- To develop the ability to design and draft electrical layouts for commercial and industrial installations.
- To provide knowledge and skills in designing substations and associated control circuits using AutoCAD.
- To enable students to apply design principles and documentation standards in a real-world electrical project using AutoCAD.

Module I: Introduction to AutoCAD for Electrical Design Overview of AutoCAD interface and tools

- Layers, blocks, and annotation in AutoCAD
- Electrical symbols: IEC/ANSI/IS standards
- Drawing and modifying basic electrical elements

Lab Experiments:

- Creating simple electrical circuit diagrams using AutoCAD
- Use of layers and blocks for electrical layouts

Module II: House Wiring and Lighting System Design

- Design of single-line diagrams (SLDs)
- Layout of internal wiring for residential buildings
- Load calculation and cable selection
- Earthing and protection system basics

Lab Experiments:

- Preparation of residential wiring layout
- Switchboard and lighting plan for 1BHK/2BHK house

Module III: Commercial and Industrial Electrical Layouts

- Design of power circuits and lighting for commercial buildings
- Distribution board design and component placement
- Panel board and bus bar layout

Lab Experiments:

- Design and drafting of distribution system for a small commercial building
- Electrical room layout with control panels

Module IV: Substation and Control Circuit Design

- Single-line diagram of substations
- Control circuit schematics
- Relay control and contactor wiring diagrams
- Cable routing and tray layout

Lab Experiments:

- Drawing of 11kV/440V substation SLD
- Panel wiring diagram for DOL/Star-Delta motor starter

Module V: Mini Project and Professional Practice

- Project planning, drawing standards, title block, and BOM
- Design and documentation of a small-scale electrical system
- Printing, plotting, and exporting drawings

Lab Activity:

- Mini-project: Design and documentation of electrical system for a small apartment, lab, or factory setup.

Course Outcomes: After completion of the course, the student should be able to

CO-1.Create and modify basic electrical diagrams using standard AutoCAD tools.
CO-2.Design and draft residential wiring and lighting layouts, including load and cable selection.
CO-3.Develop comprehensive electrical layouts for commercial buildings, including distribution boards and panels.
CO-4.Design substation SLDs and control wiring diagrams with proper component placement and routing.
CO-5.Independently complete a detailed electrical system design project, including all documentation and drawings.

Course Articulation Matrix:

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**1**	**2**
CO-1	1	2	3	3	3	2	2	-	-	-	2	2	1	2
CO-2	2	1	3	2	3	1	3	-	-	-	3	2	2	2
CO-3	1	2	3	3	3	2	3	-	-	-	2	2	3	3
CO-4	2	2	3	2	3	2	-	-	-	-	2	2	3	3
CO-5	3	2	2	2	3	2	2	-	-	-	2	2	3	3

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

II B.Tech. I Semester

Course Name	:	**ENVIRONMENTAL SCIENCE /IKS**				
Course Code	:	**2599304**	L	T	P	C
Course Category	:	**Laboratory**	1	0	0	1

CIE: 00 SEE: 100 Total: 100

INDIAN KNOWLEDGE SYSTEM

UNIT-I:

Introduction to Indian Knowledge Systems: Meaning, Nature, Scope and Salient Aspects of Bharatiya Jnana Parampara - Introduction to Vedas, Upanishads, Vidya, Kala, Jnana, Shastra - Practices and Continuity of Tradition.

Concept of Wellness – Yoga System - Ayurveda System - Ancient Indian Aesthetics

UNIT-II:

Overview of History of Indian Education and Scientific Literature: Gurukul System - Role of Sanskrit in Natural Language Processing - Scientific Literature – Vedic Literature - Available Scientific Treatises - Interlinking's

UNIT-III:

Development of Science & Technology in India: Superposition theorem, Thevenin's theorem, Norton's theorems (AC & DC), Maximum power transfer theorem, Tellegen's theorem, Compensation theorem, Millman's theorem and Reciprocity theorem (DC only).

1. Science and technology heritage of India
2. Science and technology heritage: Literary Sources
3. Science and technology heritage: Physical structures
4. Science and technology heritage: Temples
5. Science and technology heritage: Watershed management
6. Dyes, arts and perfume production.
7. Shipbuilding
8. Status of Indigenous S & T

Suggested Readings:

1. B. Mahadevan, Bhat Vinayak and Nagendra Pavan R.N., (2022) 'Introduction to Indian Knowledge Systems: Concepts and Applications' PHI learning PVT, New Delhi
2. Dharmapal (1971) 'Indian Science and Technology in the Eighteenth Century'. Other India Press, Goa.
3. Kapil Kapoor, Singh Avdhesh Kumar, (2005) 'Indian Knowledge Systems' D.K. Printworld (P)Ltd. Chakradeo, Ujwala, Temples of Bharat, Aayu Publications, New Delhi, 2024.
4. D.N. Bose, S.N. Sen and B. V. Subbarayappa, A Concise History of Science in India, Indian National Science Academy, New Delhi, 2009.
5. Datta B. and A. N. Singh, History of Hindu Mathematics: Parts I and II, Asia Publishing House, Bombay, 1962.
6. Kapoor, K. (2021), Indian Knowledge System: Nature, Philosophy, Character in Indian Knowledge System, vol. 1, Pub. Indian Institute of Advanced Studies, Shimla
7. Mahadevan, B., Bhat, V.R., Pavana, N. (2022), Philosophical Systems, in Introduction to Indian Knowledge System, Pub. PHI Learning, New Delhi.
8. Mahadevan, B., Bhat, V.R., Pavana, N. (2022), Knowledge: Framework and Classification, in Introduction to Indian Knowledge System, Pub. PHI Learning, New Delhi.

ENVIRONMENTAL SCIENCE

UNIT – IV:

Ecosystems & Natural Resources: Definition, Scope, and Importance of ecosystem. Classification, structure, and function of an ecosystem, Food chains, food webs, and ecological pyramids. Flow of energy, Bioaccumulation, ecosystem value.

Classification of Resources: Living and Non-Living resources, water resources: use and over utilization of surface and ground water, floods and droughts, Dams: benefits and problems. Mineral resources: use and exploitation, environmental effects of extracting and using mineral resources, Land resources: Forest resources, Energy resources: growing energy needs, renewable and non-renewable energy sources, use of alternate energy source.

UNIT – V:

Environmental Pollution and Control Technologies: Environmental Pollution: Classification of pollution, Ambient air quality standards.

Overview of air pollution control technologies, Concepts of bioremediation. Global Environmental Issues and Global Efforts: Climate change and impacts on human environment. Ozone depletion and Ozone depleting substances (ODS). Deforestation and desertification. International conventions / Protocols: Earth Summit, Kyoto Protocol, and Montréal Protocol. NAPCC-GoI Initiatives. Sustainable development goals (SDGs); Global environmental challenges; Environmental policies and international agreements.

Text Books:

1. Introduction to Environmental Science by Y. Anjaneyulu, BS. Publications.
2. Textbook of Environmental Studies for Undergraduate Courses by Erach Bharucha for University Grants Commission.
3. Environmental Studies by R. Rajagopalan, Oxford University Press.

Reference Books:

1. Environmental Science: towards a sustainable future by Richard T. Wright. 2008 PHL Learning Private Ltd. New Delhi.
2. Environmental Engineering and science by Gilbert M. Masters and Wendell P. Ela. 2008 PHI Learning Pvt. Ltd.
3. Environmental Science by Daniel B. Botkin & Edward A. Keller, Wiley INDIA edition.
4. Environmental Studies by Anubha Kaushik, 4th Edition, New age international publishers.
5. Text book of Environmental Science and Technology - Dr. M. Anji Reddy 2007, BS Publications.

II Year II Semester

II B.Tech. II Semester

Course Name	:	**NUMERICAL METHODS AND COMPLEX VARIABLES**				
Course Code	:	**2599402**	L	T	P	C
Course Category	:	Theory	3	0	0	3

CIE: 40 SEE: 60 Total: 100

Course Objectives: To learn

- Expressing periodic function by Fourier series and a non-periodic function by Fourier transforms
- Various numerical methods to find roots of polynomial and transcendental equations. Concept of finite differences and to estimate the value for the given data using interpolation.
- Evaluation of integrals using numerical techniques. Solving ordinary differential equations of first order using numerical techniques.
- Differentiation of complex valued functions.
- Evaluation of integrals using Cauchy's integral formula and Cauchy's residue theorem. Expansion of complex functions using Laurent's series.

UNIT-I: Fourier Series & Fourier Transforms **10 Hrs**

Fourier series: Dirichlet's Conditions – Half-range Fourier series

Fourier Transforms: Fourier Integral Theorem (Only statements), Fourier Sine and Cosine transforms (Elementary illustrations)

UNIT-II: Numerical Methods-I **10Hrs**

Solution of polynomial and transcendental equations: Bisection method – Iteration Method – Newton- Raphson method and Regula-Falsi method.

Finite differences: forward differences – backward differences – Interpolation using Newton's forward and backward difference formulae – Lagrange's method of interpolation.

UNIT-III: Numerical Methods-II **10 Hrs**

Numerical integration: Trapezoidal rule - Simpson's 1/3rd and 3/8th rules.

Ordinary differential equations: Taylor's series – Euler's method – Runge-Kutta method of fourth order for first order ODE.

UNIT-IV: Complex Differentiation **10 Hrs**

Differentiation of Complex functions – Analyticity – Cauchy-Riemann equations (without proof) – Harmonic Functions – Finding harmonic conjugate – Milne-Thomson method – Elementary analytic functions (exponential, trigonometric, logarithm).

UNIT-V: Complex Integration **10 Hrs**

Line integral – Cauchy's theorem – Cauchy's Integral formula – Zeros of analytic functions – Singularities – Laurent's series. Residues – Cauchy Residue theorem (All theorems without Proof).

Text Books:

1. B.S. Grewal, Higher Engineering Mathematics, Khanna Publishers, 36th Edition, 2010.
2. S.S. Sastry, Introductory methods of numerical analysis, PHI, 4th Edition, 2005.

References:

1. Murray R. Spiegel, Ph.D., Seymour Lipschutz, Ph.D., John J. Schiller, Ph.D., Dennis Spellman, Ph.D., Complex Variables (Schaum's outline).
2. M. K. Jain, S.R.K. Iyengar, R.K. Jain, Numerical methods for Scientific and Engineering Computations, New Age International publishers.
3. Erwin Kreyszig, Advanced Engineering Mathematics, 9thEdition, John Wiley &Sons,2006.
4. J. W. Brown and R. V. Churchill, Complex Variables and Applications, 7th Edition, Mc Graw Hill, 2004.

Course Outcomes: After learning the contents of this paper, the student must be able to

CO-1. Express any periodic function in terms of sine and cosine.

CO-2. Find the root of a given polynomial and transcendental equations. Estimate the value for the given data using interpolation.

CO-3. Find the numerical solutions for a given first order ODE's

CO-4. Analyze the complex function with reference to their analyticity, integration using Cauchy's integral and residue theorems.

CO-5. Find the integrals using Cauchy's integral formula and Cauchy's residue theorem. Expansions of Laurent's series in complex function.

Course Articulation Matrix:

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**1**	**2**
CO-1	3	2	3	2	3	2	2	2	-	-	2	2	2	2
CO-2	3	2	3	2	3	2	2	2	-	-	2	2	2	3
CO-3	3	2	3	2	3	2	2	2	-	-	2	2	2	3
CO-4	3	2	3	2	3	2	2	2	-	-	2	2	3	3
CO-5	3	2	3	2	3	2	2	2	-	-	2	2	3	3

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using CO-1. ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

II B.Tech. II Semester

Course Name	:	**ELECTRICAL MACHINES-II**				
Course Code	:	**2502401**	L	T	P	C
Course Category	:	**Theory**	3	0	0	3

CIE: 40 SEE: 60 Total: 100

Course Objectives:

- To deal with the detailed working of Three Phase induction motors
- To analyze performance of Three Phase induction motors.
- To deal with the working and characteristics of synchronous generators.
- To deal with the voltage regulation & parallel operation of alternator
- To deal with the performance and working of synchronous & single phase motors

UNIT-I: **12 Hrs**

Three Phase Induction Machines: Constructional details of cage and wound rotor machines-production of a rotating magnetic field - principle of operation - rotor EMF and rotor frequency - rotor reactance, rotor current and Power factor at standstill and during operation. Rotor power input, rotor copper loss and mechanical power developed and their inter relation. Torque equation-expressions for maximum torque and starting torque – torque-slip characteristics.

UNIT-II: **12 Hrs**

Characteristics of Induction Machines: Equivalent circuit - phasor diagram - crawling and cogging, No-load Test and Blocked rotor test –Predetermination of performance-Methods of starting and starting current and Torque calculations, Applications.

Speed Control Methods: Change of voltage, change of frequency, voltage/frequency, injection of an EMF into rotor circuit (qualitative treatment only)-induction generator-principle of operation.

UNIT-III: **13 Hrs**

Synchronous Generator (Alternator): Constructional Features of round rotor and salient pole machines – Armature windings – Integral slot and fractional slot windings; Distributed and concentrated windings – distribution, pitch and winding factors – EMF Equation. Harmonics in generated EMF – suppression of harmonics – armature reaction - leakage reactance – synchronous reactance and impedance – phasor diagram – load characteristics.

UNIT-IV: **13 Hrs**

Regulation of Synchronous Generator: Synchronous impedance method, MMF method, ZPF method and ASA methods – two reaction theory– Determination of Xd and Xq (Slip test) Phasor diagrams – Regulation of salient pole alternators.

Parallel Operation of Synchronous Generator: Synchronizing Alternators with infinite bus bars – synchronizing power torque – parallel operation and load sharing - Effect of change of excitation and mechanical power input.

UNIT-V: **11 Hrs**

Synchronous Motors: Theory of operation – phasor diagram – Variation of current and power factor with excitation – synchronous condenser – Mathematical analysis for power developed. Hunting and its suppression – Methods of starting.

Single Phase Machines: Single phase induction motor – Constructional Features-Double revolving field theory – split-phase motors – AC series motor- Universal Motor- Shaded pole motor and Applications.

Textbooks:

1. Theory and performance of Electrical machines – J.B Gupta, S.K Kataria & Sons publishers, 2009.
2. Electric Machinery- P.S. Bimbra, Khanna Publishers, 7th edition, 2010

References:

1. Electrical Machines - I.J. Nagrath& Kothari, McGraw Hill Companies, 3rd edition, 2004.
2. A Text book of Electrical Technology – B.L. Thereja and A.K. Theraja, S. ChandPublications, Vol2 eprint 2015.
3. Electrical Machines – S.K. BhattaCharya, McGraw Hill Companies, 2007.
4. Electric Machines – M.V. Deeshpande, Wheeler Publishing, 1997.

Course Outcomes:

After successful completion of the course, the student will be able to:

CO-1. Explain the construction, working principle, torque equation, and equivalent circuit of a three-phase induction motor.
CO-2. Analyze the performance characteristics, speed control methods, and industrial applications of three-phase induction motors.
CO-3. Describe the detailed working of synchronous generators.
CO-4. Demonstrate the parallel operation and evaluate the regulation of alternators.
CO-5. Illustrate the operating principles and applications of synchronous motors and single-phase induction motors.

Course Articulation Matrix:

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**1**	**2**
CO-1	1	-	3	-	2	-	-	-	-	-	1	1	1	-
CO-2	1	-	3	-	2	-	-	-	-	-	1	1	-	1
CO-3	1	-	3	-	2	-	-	-	-	-	1	1	-	1
CO-4	1	-	3	-	2	-	-	-	-	-	1	1	2	-
CO-5	1	-	3	-	2	-	-	-	-	-	1	1	2	-

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

II B.Tech. II Semester

Course Name	:	**POWER SYSTEM–II**				
Course Code	:	**2502402**	L	T	P	C
Course Category	:	**Theory**	3	0	0	3

CIE: 40 SEE: 60 Total: 100

Course Objectives:

- To know about overhead line parameters and insulators.
- To study the performance of transmission lines.
- To understand the concept of voltage control, compensation methods
- To understand the concept of per unit representation of power systems and knowledge about the power system transients.
- To know the methods of Symmetrical components and fault calculation analysis.

UNIT-I: **13 Hrs**

Overhead Transmission Lines: Line conductors, Composite conductor's transposition, bundled conductors, Inductance and capacitance of single phase and three phase lines with symmetrical spacing, and effect of earth on capacitance, skin and proximity effects.

Overhead Line Insulators: Introduction, types of insulators, Potential distribution over a string of suspension insulators, Methods of equalizing the potential, testing of insulators, Sag and Tension calculations.

UNIT-II: **12 Hrs**

Performance of Lines: Representation of lines, short transmission lines, medium length lines, nominal T and PI- representations, long transmission lines. The equivalent circuit representation of a long Line, A, B, C, and D constants, Ferranti Effect.

Corona: Introduction, disruptive critical voltage, corona loss, Factors affecting corona loss and methods of reducing corona loss, Advantages and Disadvantages of corona, interference between power and Communication lines.

UNIT-III: **12 Hrs**

Voltage Control & Power Factor Improvement: Introduction – methods of voltage control, shunt and series capacitors / Inductors, tap changing transformers, synchronous phase modifiers, power factor improvement methods.

Compensation in Power Systems: Introduction - Concepts of Load compensation – Load ability characteristics of overhead lines – Uncompensated transmission line – Symmetrical line.

UNIT-IV: **12 Hrs**

Per Unit Representation of Power Systems: The one-line diagram, impedance and reactance diagrams, per unit quantities, changing the base of per unit quantities, advantages of per unit system.

Travelling Waves on Transmission Lines: Production of travelling waves, open circuited line, short-circuited line, line terminated through a resistance, line connected to a cable, Reflection and Refraction coefficients.

UNIT-V: **13 Hrs**

Symmetrical Components and Fault Calculations: Significance of positive, negative and zero sequence components, Average 3-phase power in terms of symmetrical components, sequence impedances and sequence networks, fault calculations, sequence network equations, single line to ground fault, line to line fault, double line to ground fault, three phase fault, faults on power systems, faults with fault impedance, reactors and their location, short circuit capacity of a bus.

Textbooks:

1. C. L. Wadhwa, “Generation, Transmission and Distribution of Electrical Power”, 3rd Edition, 2017.
2. D.P. Kothari and I.J. Nagrath, “Modern Power System Analysis”, Tata Mc Graw Hill Pub. Co., New Delhi, Fifth edition, 2022.

References:

1. Chakrabarti, M.L. Soni, P.V. Gupta, U.S. Bhatnagar, “A Text book on Power System Engineering”, Dhanpat Rai Publishing Company (P) Ltd, paper back edition,2016.
2. W. D. Stevenson, “Elements of Power System Analysis”, 4th Edition, McGraw Hill, 2017.
3. Hadi Saadat, “Power System Analysis”,PSA Publishing LLC,3rd Edition, 2011.

Course Outcomes: After completion of the course, the student should be able to

CO-1: Calculate Overhead line parameters and importance of overhead insulators.
CO-2: Analyze performance of transmission lines and concept of corona.
CO-3: Understand the concept of voltage control, compensation methods.
CO-4: Understand the application of per unit quantities in power systems and analyze of travelling waves.
CO-5: Determine the fault currents for symmetrical and unbalanced faults.

Course Articulation Matrix:

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**1**	**2**
CO-1	3	-	2	-	2	-	-	-	1	-	2	-	3	2
CO-2	3	2	-	-	3	-	-	1	1	1	2	-	3	2
CO-3	3	1	2	-	2	-	-	1	2	-	2	-	3	2
CO-4	3	2	-	-	3	-	-	-	1	-	3	-	3	3
CO-5	3	2	-	-	3	-	-	-	-	-	-	-	-	-

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

II B.Tech. II Semester

Course Name	:	**DIGITAL ELECTRONICS**				
Course Code	:	**2504401**	L	T	P	C
Course Category	:	**Theory**	3	0	0	3

CIE: 40 SEE: 60 Total: 100

Course Objectives:

- To learn fundamental concepts of digital system design and common forms of number representations and their conversions.
- To implement and design logical operations using combinational logic circuits and sequential logic circuits.
- To be able to convert between different code converters.
- To understand the operations of different types of flip-flops and counters.
- To learn about Semiconductor Memories and Programmable Logic Devices

UNIT-I: 12Hrs

Fundamentals of Digital Systems and Logic Families: Digital signals, Digital circuits, AND, OR, NOT, NAND, NOR and Exclusive-OR operations, Boolean algebra, Examples of IC gates, Number systems-binary, Signed binary, Octal hexadecimal number, Binary arithmetic, One'sand Two's complements arithmetic.

UNIT-II: 12Hrs

Combinational Circuits-I: Standard representation for logic functions, K-map representation and simplification of logic functions using K- map, Minimization of logical functions, Don't care conditions, Multiplexer, De-Multiplexer

UNIT-III: 12Hrs

Combinational Circuits-II: Adders, Subtractors, Carry look ahead adder, Digital comparator, Parity checker/generator, Code converters, Priority encoders, Decoders/Drivers for display devices, Q-M method of function realization**.**

UNIT-IV: 12Hrs

Sequential Circuits: Introduction to flip-flops, SR, JK, T and D type's flip-flops, Shift registers, Conversion of flip-flops, Ring counter, Ripple (Asynchronous) counters, Synchronous counters.

UNIT-V: 12Hrs

Semiconductor Memories and Programmable Logic Devices: Memory organization and operation, expanding memory size, classification and characteristics of memories, sequential memory, read-only memory (ROM), ROM types, Read and write memory (RAM) types, Programmable logic array, Programmable array logic, Field Programmable Gate Array (FPGA).

Textbooks:

1. Kumar, "Fundamentals of Digital Circuits", Prentice Hall India, 2016.
2. M. M. Mano, "Digital logic and Computer design", Pearson Education India, 2016.

References:

1. R.S. Sedha, "A Textbook of Digital Electronics", S. Chand, 2005
2. R. P. Jain, "Modern Digital Electronics", McGraw Hill Education, 2009

Course Outcomes: After completion of the course, the student should be able to

CO1: Understand the working of logic families and logic gates.
CO2: Design and implement Combinational and Sequential logic circuits.
CO3: Implement the conversion between converters.
CO4: Design logical circuits using different flip flops.
CO5: To learn about Semiconductor Memories and Programmable Logic Devices.

Course Articulation Matrix:

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**1**	**2**
CO-1	2	2	3	3	1	2	3	3	1	1	1	1	2	2
CO-2	2	2	1	2	3	1	2	2	2	1	1	1	2	2
CO-3	3	2	2	2	2	2	2	1	1	1	1	1	2	1
CO-4	3	2	2	3	3	2	3	2	2	1	2	1	1	2
CO-5	2	3	2	2	2	1	2	1	2	1	3	1	2	2

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

II B.Tech. II Semester

Course Name	:	**CONTROL SYSTEMS**				
Course Code	:	**2502403**	L	T	P	C
Course Category	:	**Theory**	3	0	0	3

CIE: 40 SEE: 60 Total: 100

Course Objectives:

- To introduce the fundamental concepts of control systems and develop mathematical models of physical systems using transfer functions, block diagrams, and signal flow graphs.
- To analyze system dynamics in the time domain and determine system stability using classical methods.
- To develop an understanding of system behavior in the frequency domain and apply stability analysis techniques using frequency response tools.
- To develop an understanding of system behavior in the frequency domain and apply stability analysis techniques using frequency response tools.
- To present modern control system analysis using state-space models and study the internal system behavior

UNIT-I: **13 Hrs**

Mathematical modelling of physical systems: Open – loop and Closed loop Systems, Concept of Feedback Control, Benefits of Feedback and Effects of feedback, Linear, Non-Linear, Time Variant and Time Invariant systems, Mechanical and Electrical Systems. Transfer function, Block-Diagram Techniques, Signal flow graph, Controller Components: DC Servo motors, AC Servomotors, Synchro's.

UNIT-II: **12 Hrs**

Time-Domain Analysis with Input-Output Models: Time response of first and second order systems for standard test inputs. Analysis of standard Second order systems with step input, Types of System, Error Analysis for Linear time Invariant Systems, Design specifications for second-order systems based on the time- response.

Concept of Stability: Routh-Hurwitz Criteria. Relative Stability analysis, Root-Locus technique: Construction of Root-loci.

UNIT-III: **13 Hrs**

Frequency Domain Analysis: Introduction to frequency response, Relationship between time and frequency response, Concept of Bode plots and construction. Polar plots, Nyquist stability criterion. Relative stability using Nyquist criterion – gain and phase margin.

UNIT-IV: **11 Hrs**

Classical Controllers and Compensators: Proportional, Integral and Derivative Controllers- PI, PD and PID controllers, Lead, Lag and Lead-Lag compensators (elementary treatment only).

UNIT-V: **12 Hrs**

State Variable Analysis: Concept of State, State variables and State model. State Representation, Transformation of State variables, Solution of state equations and Complete response of the Systems. Concept of controllability and observability.

Textbooks:

1. I.J.Nagrath and M.Gopal, "Control Systems Engineering", New Age International, 2009.
2. B.C.Kuo, "Automatic Control System", Prentice Hall, 1995.
3. Norman S Nise, "Control Systems Engineering", Wiley, 8th Edition, 2019.

References:

1. K.Ogata, “Modern Control Engineering”, Prentice Hall, 1991.
2. K.R.Varmah, “Control Systems”, McGraw Hill Education, 2010.

Course Outcomes: After completion of the course, the student should be able to

CO-1.Model and represent mechanical and electrical systems using block diagrams and signal flow graphs and understand the impact of feedback in control systems.
CO-2.Evaluate time-domain responses of first and second-order systems and assess system stability using Routh-Hurwitz and Root Locus methods
CO-3.Construct Bode, Nyquist, and polar plots, and determine system stability using frequency domain criteria including gain and phase margins.
CO-4.Design and analyze basic controllers and compensators for system performance enhancement.
CO-5.Represent systems in state-space form, and determine controllability and observability of dynamic systems.

Course Articulation Matrix:

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**1**	**2**
CO-1	3	2	2	2	3	-	1	1	2	-	3	1	3	2
CO-2	3	3	2	-	3	-	1	1	2	-	3	1	3	2
CO-3	3	3	2	-	3	-	1	1	2	-	3	1	3	3
CO-4	3	2	3	2	3	1	2	2	2	1	3	2	3	3
CO-5	3	2	2	1	3	-	1	1	2	-	3	1	3	3

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

II B.Tech. II Semester

Course Name	:	**COMPUTATIONAL MATHEMATICS LABORATORY (Using Python/MATLAB Software)**				
Course Code	:	**2599403**	L	T	P	C
Course Category	:	**Laboratory**	0	0	2	1

CIE: 40 SEE: 60 Total: 100

Course Objectives:

To learn

- Solve problems of Eigen values and Eigen Vectors using Python/MATLAB.
- Solution of Algebraic and Transcendental Equations using Python/MATLAB
- Solve problems of Linear system of equations
- Solve problems of First-Order ODEs and applications
- Solve problems of Higher order ODEs with constant coefficients

UNIT-I: Eigen values and Eigenvectors: **6P**

Programs:

- Finding real and complex Eigen values.
- Finding Eigen vectors

UNIT-II: Solution of Algebraic and Transcendental Equations: **6P**

Bisection method, Newton Raphson Method

Programs:

- Root of a given equation using Bisection method.
- Root of a given equation Newton Raphson Method.

UNIT-III: Linear system of equations **6P**

Jacobi's iteration method and Gauss-Seidal iteration method

Programs:

- Solution of given system of linear equations using Jacobi's method
- Solution of given system of linear equations using Gauss-Seidal method

UNIT-IV: First-Order ODEs **6P**

Exact and non-exact equations, Applications: exponential growth/decay, Newton's law of cooling.

Programs:

- Solving exact and non-exact equations
- Solving exponential growth/decay and Newton's law of cooling problems

UNIT-V: Higher order linear differential equations with constant coefficients: **6P**

Programs:

- Solving homogeneous ODEs
- Solving non-homogeneous ODEs

Textbooks:

1. MATLAB and its Applications in Engineering, Rajkumar Basal, Ashok Kumar Geo, Manoj Kumar Sharma, Pearson publication.
2. Kenneth A. Lambert, The fundamentals of Python: First Programs, 2011, Cengage Learnings.
3. Think Python First Edition, by Allen B. Downey, Orielly publishing.
4. Introduction to Python Programming, William Mitchell, Povel Solin, Martin Novak et al., NC Lab Public Computing, 2012.
5. Introduction to Python Programming, ©Jacob Fredslund, 2007.

References:

1. An Introduction to Python, John C. Lusth, The University of Alabama, 2011.
2. Introduction to Python, ©Dave Kuhlman, 2008.

Course Outcomes: After completion of the course, the student should be able to

CO-1. Develop the code to find the Eigen values and Eigen Vectors using Python/MATLAB.
CO-2. Develop the code find solution of Algebraic and Transcendental Equations and Linear system of equations using Python/MATLAB
CO-3. Write the code find the system of linear equations using Jacobi's method, Gauss-Seidal method.
CO-4. Write the code to solve problems of First-Order ODEs equations with constant coefficients
CO-5. Write the code to solve problems of homogeneous & non-homogeneous ODEs

- Visualize all solutions Graphically through programmes

Course Articulation Matrix:

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO-1	1	2	3	3	3	2	3	-	-	-	2	2	2	2
CO-2	2	2	3	3	3	2	3	-	-	-	2	2	3	3
CO-3	1	2	3	3	3	2	3	-	-	-	2	2	3	3
CO-4	2	2	3	2	3	2	-	-	-	-	2	2	3	3
CO-5	3	2	2	2	3	2	2	-	-	-	2	2	3	3

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

II B.Tech. II Semester

Course Name	:	**ELECTRICAL MACHINES-II LABORATORY**				
Course Code	:	**2502404**	L	T	P	C
Course Category	:	**Laboratory**	0	0	2	1

CIE: 40 SEE: 60 Total: 100

Course Objectives:

- To train the students to perform tests on transformer
- To train the students to perform tests and analyze induction motor
- To train the students to perform tests on alternator.
- To train the students to analyze alternator
- Train the students to perform tests and analyze synchronous motor.

The following experiments are required to be conducted as compulsory experiments:

1. Sumpner's test on a pair of single-phase transformers
2. No-load & Blocked rotor tests on three phase Induction motor
3. Regulation of a three–phase alternator by synchronous impedance & MMF methods
4. 'V' and 'Inverted V' curves of a three—phase synchronous motor.
5. Equivalent Circuit of a single-phase induction motor
6. Determination of X_d and X_q of a salient pole synchronous machine
7. Brake Test on three phase Induction Motor
8. Regulation of three-phase alternator by ZPF and ASA methods

In addition to the above experiments, atleast any two of the following experiments are required to be conducted from the following list:

1. Separation of core losses of a single-phase transformer
2. Efficiency of a three-phase alternator
3. Parallel operation of Single-phase Transformers
4. Measurement of sequence impedance of a three-phase alternator.
5. Scott Connection of transformer

Course Outcomes: After completion of the course, the student should be able to

CO-1. Test and analyze the performance of transformer.
CO-2. Test and analyze the performance of induction motor
CO-3. Test the performance of alternator.
CO-4. Analyze the performance of alternator
CO-5. Test and analyze the performance of synchronous motor.

Course Articulation Matrix:

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**1**	**2**
CO-1	1	3	3	-	2	-	-	-	-	-	1	1	1	-
CO-2	1	-	-	-	2	-	-	-	-	-	1	1	-	2
CO-3	1	3	3	-	2	-	-	-	-	-	1	1	3	-
CO-4	1	3	3	-	2	-	-	-	-	-	1	1	2	-
CO-5	1	3	3	-	2	-	-	-	-	-	1	1	2	-

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

II B.Tech. II Semester

Course Name	:	**CONTROL SYSTEMS LABORATORY**				
Course Code	:	**2502405**	L	T	P	C
Course Category	:	**Laboratory**	0	0	2	1

CIE: 40 SEE: 60 Total: 100

Course Objectives:

- Provide practical exposure to analyze the time and frequency response of first- and second-order systems.
- Familiarize students with the working principles and characteristics of control components such as synchros, AC/DC motors, and compensators.
- Develop the ability to model, simulate, and analyze control systems using modern software tools.
- Enable students to design, implement, and evaluate controllers (P, PI, PD, PID) for system performance improvement.
- Impart hands-on experience in state-space representation, stability analysis, and controller design for real-time applications.

The following experiments are required to be conducted compulsory experiments:

1. Time response of Second order system
2. Characteristics of Synchro's
3. Characteristics of AC servo motor
4. Programmable logic controller– Study and verification of truth tables of logic gates, simple Boolean expressions.
5. Transfer function of DC generator
6. Lag and lead compensation–Magnitude and phase plot
7. Stability analysis (Bode, Root Locus, Nyquist) of Linear Time Invariant system using simulation tools.
8. State space model for classical transfer function using simulation tools.

In addition to the above eight experiments, at least any two of the experiments from the following list are required to be conducted:

1. Temperature controller using PID
2. Effect of P, PD, PI, PID Controller on a second order systems
3. (a) Simulation of P, PI, PID Controller.
   (b) Linear system analysis (Time domain analysis, Error analysis) using suitable software
4. Transfer function of DC motor
5. Design of Lead-Lag compensator for the given system and with specification using suitable software.

Course Outcomes: After completion of the course, the student should be able to

CO-1.Analyze and interpret the time response of second-order systems and assess performance parameters.

CO-2.Examine the characteristics of synchros, AC servo motors, and DC machines for control applications.

CO-3.Apply simulation tools to determine system stability using Bode, Nyquist, and Root Locus techniques.

CO-4. Design and analyze compensators and controllers (P, PI, PD, PID) for performance enhancement.

CO-5.Represent dynamic systems in state-space form and validate controllability and observability using software tools.

Course Articulation Matrix:

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**1**	**2**
CO-1	3	3	2	1	3	-	1	1	2	-	3	1	3	2
CO-2	3	2	2	1	3	-	1	1	2	-	3	1	3	2
CO-3	3	3	2	1	3	-	1	1	2	-	3	1	3	3
CO-4	3	2	3	2	2	1	2	2	2	1	3	2	3	3
CO-5	3	2	2	1	1	-	1	1	2	-	3	1	3	3

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

II B.Tech. II Semester

Course Name	:	**DIGITAL ELECTRONICS LABORATORY**				
Course Code	:	**2504402**	L	T	P	C
Course Category	:	**Laboratory**	0	0	2	1

CIE: 40 SEE: 60 Total: 100

Course Objectives:

- To learn basic techniques for the design of digital circuits.
- To learn basic techniques for number conversion systems.
- To implement simple logical operations using combinational logic circuits.
- To design combinational logic circuits.
- To design sequential logic circuits.

List of Experiments:

1. Realization of Boolean Expressions using Gates
2. Design and realization logic gates using universal gates
3. Generation of clock using NAND/NOR gates
4. Design a 4 – bit Adder / Subtractor
5. Design and realization a 4 – bit gray to Binary and Binary to Gray Converter
6. Design and realization of a 4-bit pseudo random sequence generator using logic gates.
7. Design and realization of an 8-bit parallel load and serial out shift register using flip-flops.
8. Design and realization Asynchronous and Synchronous counters using flip-flops
9. Design and realization 8x1 using 2x1 mux
10. Design and realization 2-bit comparator
11. Verification of truth tables and excitation tables
12. Realization of logic gates using DTL, TTL, ECL, etc.,

Text Books:

1. A. Kumar, "Fundamentals of Digital Circuits", Prentice Hall India, 2016.
2. M. M. Mano, "Digital logic and Computer design", Pearson Education India, 2016.

Reference Books:

1. R.S. Sedha, “A Textbook of Digital Electronics”, S.Chand, 2005
2. R. P. Jain, "Modern Digital Electronics", McGraw Hill Education, 2009.

Course Outcomes: After completion of the course, the student should be able to

CO1: Understand the working of digital circuits.
CO2: Design and implement logic families and logic gates.
CO3: Design and implement Combinational logic circuits.
CO4: Design and implement Sequential logic circuits.
CO5: Analyze different types of semiconductor memories

Course Articulation Matrix:

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**1**	**2**
CO-1	2	2	3	3	2	1	1	3	3	3	3	-	-	2
CO-2	1	3	1	2	3	3	1	3	2	2	3	-	-	2
CO-3	1	2	1	1	3	3	1	3	3	3	3	-	-	2
CO-4	1	3	2	3	1	2	3	1	2	2	2	-	-	2
CO-5	1	2	1	2	3	1	2	3	1	2	3	-	-	2

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

II B.Tech. II Semester

Course Name	:	**PCB DESIGN**				
Course Code	:	**2502406**	L	T	P	C
Course Category	:	**Laboratory**	0	0	2	1

CIE: 40 SEE: 60 Total: 100

Course Objectives:

- To introduce students to various types of PCBs, their materials, manufacturing processes, and design standards using EDA tools.
- To develop the ability to create accurate circuit schematics and perform electrical rule checks using PCB CAD software.
- To teach component placement, footprint assignment, and routing techniques in PCB layout design.
- To familiarize students with the generation of output files like Gerbers and the basics of PCB fabrication processes.
- To provide hands-on experience in assembling, soldering, and testing a complete PCB-based electronic project.

Module I: Fundamentals of PCB Design

- Types of PCBs: Single-layer, Double-layer, Multilayer
- PCB materials and manufacturing process
- PCB design rules and standards (IPC standards)
- Introduction to EDA tools (e.g., KiCad, Eagle, Altium, EasyEDA)

Lab Activity:

- Exploring the user interface of PCB design software
- Setting up design rules

Module II: Schematic Design

- Creating circuit schematics using PCB CAD tools
- Component library management
- Electrical rule checking (ERC)
- Netlist generation

Lab Activity:

- Designing a basic power supply or LED flasher circuit
- Performing ERC and generating netlist

Module III: PCB Layout and Routing

- Importing netlist to layout editor
- Footprint assignment and component placement
- Manual vs auto-routing
- Design Rule Check (DRC)

Lab Activity:

- Placing components and routing for the schematic designed earlier
- Performing DRC and correcting errors

Module IV: PCB Output Files and Fabrication

- Generating Gerber files, drill files, and BOM
- Understanding layers (Top, Bottom, Soldermask, Silkscreen)
- PCB printing, photoresist method, and etching
- Introduction to SMD and through-hole assembly

Lab Activity:

- Generate Gerber files and preview using Gerber viewer

- Fabricate a basic single-layer PCB (simulation or actual lab process)

Module V: Mini Project and Testing

- Assembling components on fabricated PCB
- Soldering and desoldering techniques
- Continuity testing and troubleshooting
- Mini-project: Design a simple power supply, logic gate trainer, or timer circuit

Lab Activity:

- Complete mini project: From schematic to testing of PCB

Course Outcomes: After completion of the course, the student should be able to

CO-1.Identify different types of PCBs and set up design rules in EDA tools following industry standards.

CO-2.Design schematics and generate netlists while ensuring electrical correctness through ERC.

CO-3.Place components, perform routing, and conduct Design Rule Checks to create error-free PCB layouts.

CO-4.Generate and verify Gerber files, understand PCB layers, and simulate or perform basic fabrication processes.

CO-5.Assemble, solder, test, and troubleshoot a functional PCB from their own schematic and layout design.

Course Articulation Matrix:

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**1**	**2**
CO-1	2	2	2	1	3	1	2	-	-	-	2	2	2	3
CO-2	3	3	3	2	3	1	1	-	-	-	2	2	3	3
CO-3	3	3	3	2	3	1	2	-	-	-	2	2	3	3
CO-4	2	2	2	2	3	2	1	-	-	-	2	2	2	3
CO-5	3	2	3	2	3	2	2	-	-	-	3	2	3	3

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

II B.Tech. II Semester

Course Name	:	**GENDER SENSITIZATION LAB/ HUMAN VALUES AND PROFESSIONAL ETHICS**				
Course Code	:	**2599404**	L	T	P	C
Course Category	:	**Laboratory**	1	0	0	1

CIE: 00 SEE: 100 Total: 100

Course Objectives:

- To enable the student to understand the core values that shapes the ethical behaviour. And Student will be able to realize the significance of ethical human conduct and self- development
- Students will be able to inculcate positive thinking, dignity of labour and religious tolerance.
- The students will learn the rights and responsibilities as an employee and a team member.
- Students will attain a finger grasp of how gender discrimination works in our society and how to counter it.
- Students will develop a better understanding on issues related to gender and Empowering students to understand and respond to gender violence.

UNIT-I
Values And Self-Development
Social values and individual attitudes, Importance of cultivation of values, Sense of duty, Devotion, Self-reliance, Confidence, Concentration, Truthfulness, Cleanliness, Honesty, Humanity, Power of faith, National unity, Patriotism, Love for nature, Discipline.

UNIT-II
Personality And Behavior Development
Positive thinking, punctuality, avoiding fault finding, Free from anger, Dignity of labour, religious tolerance, Aware of self-destructive habits.

UNIT-III
Introduction To Professional Ethics
Basic Concepts, Governing Ethics, Personal & Professional Ethics, Ethical Dilemmas, Life Skills, Emotional Intelligence, Thoughts of Ethics, Value Education, Dimensions of Ethics, Profession and professionalism, Professional Associations, Professional Risks, Professional 90 Accountabilities, Professional Success, Ethics and Profession.

UNIT-IV
Introduction To Gender
Definition of Gender, Basic Gender Concepts and Terminology, Attitudes towards Gender, Social Construction of Gender.

UNIT-V
Gender-Based Violence
The concept of violence, Types of Gender-based violence, the relationship between gender, development and violence, Gender-based violence from a human rights perspective.

Textbooks:
1. Professional Ethics Includes Human Values (2nd Edition) By R Subramanian, Oxford University Press, 2017.
2. Ethics in Engineering Practice & Research, Caroline Whit beck, 2e, Cambridge University Press 2015.
3. A Bilingual Textbook on Gender" written by A. Suneetha, Uma Bhrugubanda, Duggirala Vasanta, Rama Melkote, Vasudha Nagaraj, Asma Rasheed, Gogu Shyamala, Deepa Sreenivas and Susie Tharu and published by Telugu Akademi, Hyderabad, Telangana State in the year 2015.

References:

1. Menon, Nivedita. Seeing like a Feminist. New Delhi: Zubaan-Penguin Books, 2012
2. Abdulali Sohaila. “I Fought For My Life…and Won.”Available online at: http://www.thealternative.in/lifestyle/i-fought-for-my-lifeand-won-sohaila-abdulal/
3. Engineering Ethics, Concepts Cases: Charles E Harris Jr., Michael S Pritchard, Michael J Rabins, 4e, Cengage learning, 2015.
4. Business Ethics concepts & Cases: Manuel G Velasquez, 6e, PHI, 2008